



Supplementary Materials for

Hydroformylation of pyrolysis oils to aldehydes and alcohols from polyolefin waste

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Materials and Methods

Materials

Colored Post-Consumer high-density polyethylene (PCR-HDPE) was obtained from a material recovery facility in Mexico, being mostly from personal care packaging.

Virgin high-density polyethylene resin (HDPE) was obtained from Amcor, the grade is Nova Surpass HPs167-AB.

Virgin Low-density polyethylene resin (LDPE) was obtained from Amcor, the grade is NA216000

Virgin polypropylene (PP) was obtained from Amcor, the grade is D218 Braskem

Chemicals below were used as purchased.

1-hexene (Sigma Aldrich, Product No. 240761, $\geq 99\%$)

cis-2-hexene (Sigma Aldrich, Product No. 538493, $\geq 99\%$)

1,5 hexene (Sigma Aldrich, Product No. 128554, $\geq 97\%$)

Heptanal (Sigma Aldrich, Product No. W254002, $\geq 95\%$)

Heptanol (Sigma Aldrich, Product No. H2805, 98%)

Octanal (Sigma Aldrich, Product No. O5608, $\geq 99\%$)

Octanol (Sigma Aldrich, Product No. 472328, $\geq 99\%$)

Nonanal (Sigma Aldrich, Product No. W278220, $\geq 98\%$)

Nonanol (Sigma Aldrich, Product No. 74280, $\geq 98\%$)

6-Methyl-1-heptanol (AstaTech Inc, Product No. AST-A10851 0.25G, 93%)

2-Ethyl-1-hexanol (Santa Cruz Biotechnology, Product No. sc-230267)

2-Propyl-1-pentanol (Santa Cruz Biotechnology, Product No. sc-230650)

Hexane (Sigma Aldrich, Product No. 675393, $\geq 95\%$)

Toluene (Sigma Aldrich, anhydrous, Product No. 244511, 99.8%)

Dodecane (Sigma Aldrich, anhydrous, Product No. 297879, $\geq 99\%$)

Water (18 M Ω Milli-Q)

Chloroform-d (CDCl₃, Sigma Aldrich, Product No. 151823, 99.8 atom %D)

2-propanol (Sigma Aldrich, Product No. 437522, $\geq 99.5\%$)

C8-C40 paraffins calibration standard (Sigma Aldrich, Product No. 40147-U)

2-ethyl-1-butene (Sigma Aldrich, Product No. E14705, $\geq 95\%$)

1,4-hexadiene, mixture of *cis* and *trans* (Sigma Aldrich, Product No. 129186, $\geq 99\%$)

3,3-dimethyl-1-butene (Sigma Aldrich, Product No. 39832, $\geq 98.5\%$)
1-methylcyclopentene (Sigma Aldrich, Product No. M39806, $\geq 98\%$)
Methylcyclopentane (Sigma Aldrich, Product No. M39407, $\geq 97\%$)
Cyclohexene (Sigma Aldrich, Product No. 125431, $\geq 99\%$)
Heptane (Sigma Aldrich, Product No. 34873, $\geq 99\%$)
3,3-dimethyl-1-pentene (Sigma Aldrich, Product No. R289280)
2-ethyl-1-hexene (Sigma Aldrich, Product No. CDS000215)
2-methyl-1-hexene (Sigma Aldrich, Product No. 111627, $\geq 96\%$)
4-methyl-1-hexene (Sigma Aldrich, Product No. 67500, $\geq 98\%$)
2-methylheptane (Sigma Aldrich, Product No. M47949, $\geq 98\%$)
1-octene (Sigma Aldrich, Product No. 74900, $\geq 99.5\%$)
1,7-octadiene (Sigma Aldrich, Product No. O2501, $\geq 98\%$)
1-nonene (Sigma Aldrich, Product No. 74323, $\geq 99.5\%$)
1-decene (Sigma Aldrich, Product No. 30650, $\geq 97\%$)
1,9-decadiene (Sigma Aldrich, Product No. 118303, $\geq 97\%$)
Decan (Sigma Aldrich, Product No. 457116, $\geq 99\%$)
1-undecene (Sigma Aldrich, Product No. 242527, $\geq 97\%$)
1-dodecene (Sigma Aldrich, Product No. 44148, $\geq 99\%$)
1-tetradecene (Sigma Aldrich, Product No. 87189, $\geq 97\%$)
1-pentadecene (Sigma Aldrich, Product No. 222887, $\geq 98\%$)
1-heptadecene (Sigma Aldrich, Product No. H1108, $\geq 98\%$)
Benzene (Sigma Aldrich, Product No. 12450, $\geq 99.9\%$)
P-xylene (Sigma Aldrich, Product No. 296333, $\geq 99\%$)
M-xylene (Sigma Aldrich, Product No. 296325, $\geq 99\%$)
O-xylene (Sigma Aldrich, Product No. 95662, $\geq 99\%$)
Ethylbenzene (Sigma Aldrich, Product No. 296848, $\geq 99.8\%$)
Styrene (Sigma Aldrich, Product No. S4972, $> 99\%$)
Propylbenzene (Sigma Aldrich, Product No. P52407, $> 98\%$)
Decahydronaphthalene (Sigma Aldrich, Product No. 8031010100, $> 99\%$)
Naphthalene (Sigma Aldrich, Product No. 147141, $> 99\%$)
1-methylnaphthalene (Sigma Aldrich, Product No. W319309, $> 95\%$)

Co₂(CO)₈ (Sigma Aldrich, Product No. 60811, ≥90%Co)

1,2-Bis(diphenylphosphino) ethane monoxide (dpe, Santa Cruz Biotechnology, Product Code SC-222796)

Tri-n-butylphosphine (Bu₃P, Sigma Aldrich, Product No. 731374, Batch SHBN3708)

Acetic acid (Sigma Aldrich, ACS reagent, Product No. A6283, ≥99.7%)

Co(NO₃)₂·6H₂O (Sigma Aldrich, Product No. 239267, ≥98%)

Ni(NO₃)₂·6H₂O (Sigma Aldrich, ACS reagent, Product No. 203874, 99.999% trace metal basis)

SiO₂ (Sigma Aldrich, Product No. 236845, Davisil Grade 646)

5 wt% Ru/C (Sigma Aldrich, Product No. 206180)

Zr(OH)₄ (Sigma Aldrich, Product No. 464171, 97%)

Cu(NO₃)₂·3H₂O (Sigma Aldrich, Product No. 61194)

Methods

All experiments were repeated at least two times to ensure reproducibility. Pyrolysis of different plastics was carried out in a home-built fluidized bed reactor. The detailed procedure was described elsewhere.⁽⁶⁾ Residence time was calculated based on Equation 1.

$$\tau = \frac{\text{Volumetric gas flow rate}}{V_{\text{reactor}}} \quad (1)$$

The volumetric gas flow rate was the gas flow rate at targeted temperature and converted by ideal gas law. The V_{reactor} refers to the empty space in the reactor: the actual reactor volume minus the sand volume.

The pyrolysis oil obtained from all condensers was mixed first in a round bottom flask and then distilled to a light cut and heavy cut to simplify the characterization (**Fig. S1**). The distillation system is shown below. The round bottom flask was put in an oil bath with a stir bar and was heated up to 165 °C by a hot plate and held isothermally for 10 minutes. The round bottom flask was also connected to a cylindrical condenser which was flowed through 10 °C cold water to help condense the light oil. The light oil was collected by a round bottom flask which was connected to the cylindrical condenser through a bent adaptor and the round bottom flask was put in a dry ice container. A vacuum pump (Pfeiffer D-35614 Asslar) was connected to the adaptor. The actual boiling point of the oil cuts was around 175 °C.

1-Nonene and N-decane were chosen to be used as the model compounds to verify the DFT calculations. 1-Nonene was pyrolyzed at 500 °C with residence times of 20 s and 30 s. N-decane

at 500 °C with residence times at 20s, and 30 s, and 600 °C with residence times at 20 s, and 30 s. Since >30% conversion of 1-Nonene was achieved at 500 °C, we didn't perform the 1-Nonene pyrolysis at 600 °C.

The pyrolysis of model compounds was carried out in a fixed-bed quartz tube reactor with an outer diameter of ½ inch. Acid and base washed and, the same as the one used in the fluidized bed reactor, was loaded to the fixed-bed reactor with the same sand-to-reactor volume ratio as the fluidized bed reactor. The model compounds were fed by a syringe pump (Fisherbrand™ Single Syringe Pump) to an evaporate chamber at 200 °C. The vaporized compounds were mixed with N₂ at the evaporate chamber to reach a constant partial pressure of 16.2 kPa. All reactions proceed under ambient pressure. A bubbler was used as the condenser to collect the liquid products, which was cooled by an ice water bath (temperature: 0 °C). The liquid samples were then diluted to 25ml with ethanol for GC analysis. A gas bag was connected to the bubbler's outlet to collect the non-condensable gases, and the gas products were analyzed offline using a Refinery Gas Analyzer GC.

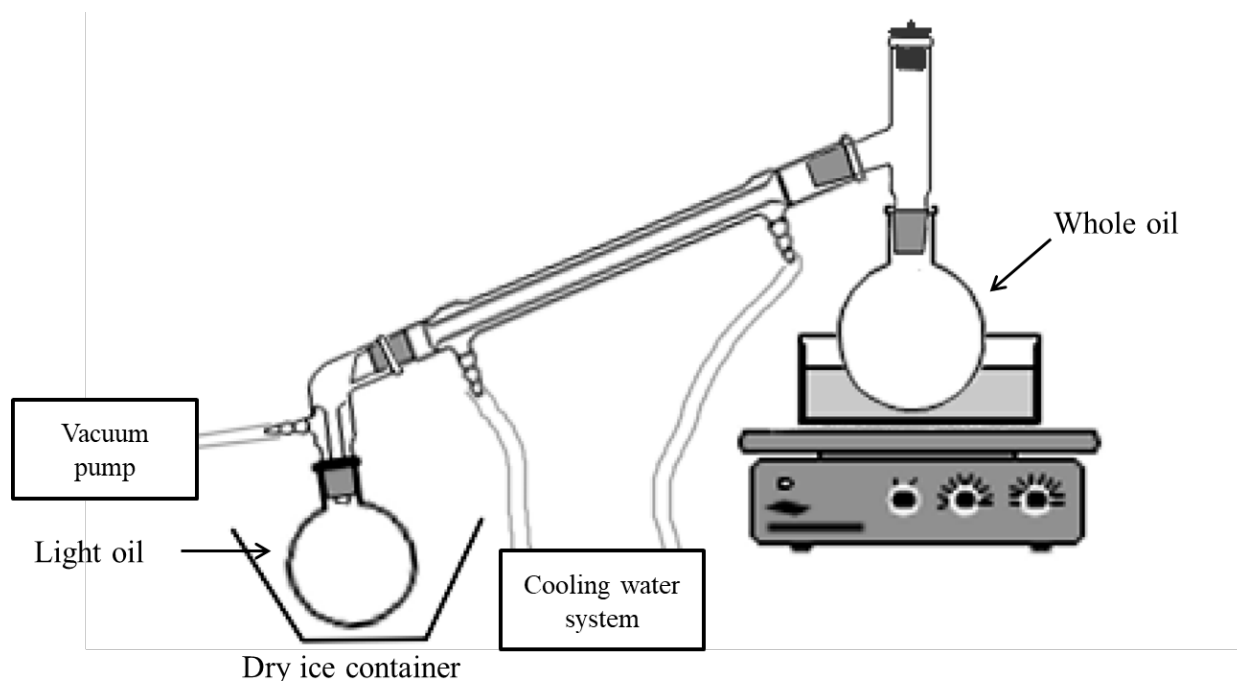


Fig. S1. Plastic oil distillation system scheme.

Co₂(CO)₈, dpe, Bu₃P, and 5 wt%Ru/C were used as purchased. 10 wt%Cu/ZrO₂ and 20 wt%Ni/SiO₂ were synthesized by wetness impregnation method. Zr(OH)₄ was heated to 500 °C (ramping rate: 4 °C /min) for 4 h in static air. The Cu was then introduced to the obtained ZrO₂ via

incipient wetness impregnation method. Both 10 wt%Cu/ZrO₂ and 20 wt%Ni/SiO₂ were calcined at 500 °C (ramping rate: 4 °C /min) for 5 h in static air after incipient wetness impregnation.

Hydroformylation of pyrolysis oil model compounds (i.e., 1-hexene, *cis*-2-hexene and 1,5-hexadiene) was performed using modified and unmodified Co catalyst. Co₂(CO)₈ (0.8 mol %) catalyst was dispersed in 90 mL toluene containing a 1.4 mol % model compound in the presence or absence of a ligand (P/Co molar ratio=1 for dpe(50), P/Co molar ratio=4 for Bu₃P(51)) in a glove box. Dodecane (0.2 wt %) was added as an internal standard. For the hydroformylation of pyrolysis oil, Co₂(CO)₈ (1/10 Co₂(CO)₈/Mixture mass ratio) was dispersed in the pyrolysis oil in a glove box. In a typical experiment, 20-40 g of plastic oil was used. The mixture was then transferred to a 350 mL stainless steel Parr reactor for the hydroformylation of the pyrolysis oil at 120 °C, 70 Bar (CO/H₂=1, molar ratio, Air Gas). The reactor was purged thrice with Ar (Air Gas) at room temperature (25 °C) then increased to 120 °C in 25 mins. Syngas (CO/H₂=1, molar ratio) was introduced to the reactor after the system reached 120 °C. The mixture was stirred at 900 rpm during the whole process. A dipping tube (1/16") was used for sampling during the reaction.

CHN(O)S elemental analysis was performed using Elementar (vario MICRO cube) elemental analyzer. The hydroformylated oil was measured before Co removal. To determine the oxygen introduced during the hydroformylation process, we subtracted the amount of Co catalyst from the obtained oxygen content and performed the calculation. To remove the Co catalyst from the hydroformylated products, the products were stirred while exposed to air in an aqueous solution containing twice the volume of aqueous solution with acetic acid (10 wt%) and Co(NO₃)₂·6H₂O (2.5 wt%). In a typical experiment, approximately 10 g of hydroformylated products were stirred in a glass container (Qorpak Glass Bottles, 125 mL) for 10 h, or until the color of organic phase turned from dark brown to light yellow, to remove Co catalyst. The organic phase, which contained the hydroformylated products, was then separated from the aqueous phase.

Hydrogenation of octanal and hydroformylated products was carried out in a 45 mL stainless steel Parr batch reactor or a continuous flow reactor. 1.5 g of 5 wt%Ru/C, 10 wt%Cu/ZrO₂, or 20 wt%Ni/SiO₂ was reduced separately in a 45 mL stainless steel Parr batch reactor with H₂ (20 Bar, 99.99%, Air Gas) for 6 h at 350 °C. The system was purged with hydrogen every 30 mins. The pretreated catalysts were stored in a glove box. For the experiments using batch reactors, the reactor was loaded and sealed in a glove box. The reactor was then transported to the Parr Multiple Reactor System, purged three times with H₂ followed by heating up to 100 °C. The reactor was

then pressurized and held for a certain time. For the experiments using continuous flow reactor, the catalyst (20 wt%Ni/SiO₂) was loaded to a 1/4" Swagelok stainless steel tube reactor followed by pretreatment at 400 °C (ramping rate: 2 °C/min) for 4 h in H₂ flow at 50 standard cubic centimeters per minute (SCCM). The reactor was then cooled down to 100 °C and pressurized to 78 Bar (H₂) by a back pressure regulator. Hydroformylated products were introduced by an HPLC pump. Heavy cut was dispersed in hexane (equal weight ratio) before feeding to the reactor. The catalyst was regenerated every 10 h time on stream by flowing air (Air Gas, 50 SCCM) at 400 °C followed by the same pretreatment with H₂.

The liquid samples from the pyrolysis of model compounds were analyzed by both GC-FID (Shimazu, Nexis GC2030) and GC-MS (Shimazu, GC 2010). Both GCs were equipped with a 30m RTX-VMS column (Restek). GC-MS was used to identify unknown compounds and GC-FID was used to quantify the products. For each measurement, the initial set temperature for the oven was 40 °C and held for 5 min. The oven was then heated to 100 °C with a ramping rate of 5 °C/min, then heated to 240 °C with a ramping rate of 20 °C/min, and finally held at 240 °C for another 11 min. C₅~C₁₀ paraffins and olefins and BTX were diluted in ethanol to calibrate the signals.

The gas products from the pyrolysis of model compounds were analyzed using a custom-modified Refinery Gas Analyzer Shimazu GC 2014 with 1) Restek RTX-alumina column with an FID detector to analyze C₁-C₆ hydrocarbons, and 2) RTX-Q-plot column and RTX-MS-5A column with a thermal conductivity detector (TCD) to quantify the generated H₂. For each measurement, the initial set temperature for the oven was 40 °C and held for 1.65 min. The oven was then heated to 90 °C with a ramping rate of 20 °C/min and held at 90 °C for 1min. Finally, the oven was heated to 150 °C and held at 150 °C for 25 min. Scott gas mixtures (1000 ppm C₁~C₆ n-paraffins, 1000 ppm C₂~C₆ 1-olefins, and 10000 ppm H₂) were used to calibrate the signals.

One-dimensional (1D) gas chromatograph (GC) Flame Ionization Detector (FID) analysis was performed using a Shimazu 2010 GC. Sample was diluted in hexane (sample/hexane mass ratio=1/9) and then analyzed by GC. The GC is equipped with a Restek MXT-1HT column (Catalog number 70132). The column length is 10 m with a 0.53 mm inner diameter. For each experiment, the initial set temperature for the oven was 35 °C and held for 5 min. The oven was then heated to 415 °C with a temperature ramping rate of 6 °C/min and held at 415 °C for another 6.15 min. Hydrogen was the carrier gas and the flow rate for hydrogen was 40 ml/min.

Two-dimensional (2D) gas chromatograph (GC×GC) FID analysis was performed with an Agilent Technologies 7890B GC. Sample was diluted in hexane (sample/hexane mass ratio=1/9) before analyzed by GC×GC-FID. Note that, the upgraded oils (hydroformylated and hydrogenated) were also diluted in 2-propanol before GC×GC-FID analysis to enhance solubility of aldehydes and alcohols present in the oil. The carbon distribution of the final product was calculated based on the GC×GC chromatogram using 2-propanol as diluent. The GC×GC-FID is equipped with a capillary Agilent 30 m VF-17ms column (CP8981) and a capillary Agilent 5 m DB-1 column (121-1011), flowed by hydrogen as the carrier gas with a flowrate at 2.5 ml/min. The inner diameter for both columns is 0.25mm. Series of chemicals were injected into GC×GC-FID for the assignment of the blobs in the 2D chromatogram. The products were first injected into a column VF-17ms, which is mid-polarity and is used to separate compounds through polarity, then injected into DB-1 column, which does not have low polarity that can be used to separate compounds through the boiling points. For each measurement, the initial set temperature for the oven was 35 °C and held for 4 min. The oven was then heated to 320 °C with a ramping rate of 4 °C/min and held at 320 °C for another 15 min. The modulator relative time was set to 1.8 s. Hydrogen was the carrier gas and the flow rate for hydrogen was 40 ml/min.

In the pyrolysis of model compounds, the conversion and product selectivity in experiments are averaged with standard deviations by Equation 2-4:

$$\text{Conversion} = \frac{n_1 - n_2}{n_1} \times 100\% \quad (2)$$

n_1 = molar flow rate of reactant in

n_2 = molar flow rate of reactant out

Here, the reactant includes both the reactant fed in and its isomers (for 1-nonene).

$$\text{Carbon Selectivity} = \frac{n_i \times \text{corresponding carbon number}}{(n_1 - n_2) \times \text{corresponding carbon number}} \times 100\% \quad (3)$$

$$\text{Hydrogen Selectivity} = \frac{n_i \times \text{corresponding hydrogen number}}{(n_1 - n_2) \times \text{corresponding hydrogen number}} \times 100\% \quad (4)$$

n_i = molar flow rate of species-i

The conversion of model compounds hydroformylation and hydrogenation was calculated using 1D GC based on Equation 5:

$$\text{Conversion} = \frac{c_1 - c_2}{c_1} \times 100 \quad (5)$$

c_1 = concentration of substrates before reaction

c_2 = concentration of substrates after reaction

Here, the substrates include both the reactant and its isomers.

The conversion of pyrolysis oil hydroformylation was calculated using GC×GC based on Equation 6:

$$\text{Conversion} = \frac{v_1 - v_2}{v_1} \times 100 \quad (6)$$

v_1 = volume of the blobs belonging to olefins in the pyrolysis oil

v_2 = volume of the blobs belonging to olefins in the hydroformylated oil

The conversion of hydrogenation of the hydroformylated oil was calculated using quantitative ^{13}C NMR based on Equation 7:

$$\text{Conversion} = \frac{a_1 - a_2}{a_1} \times 100 \quad (7)$$

a_1 = area of the peaks belonging to formyl groups in the hydroformylated oil

a_2 = area of the peaks belonging to formyl groups in the hydroformylated oil after hydrogenation

In this calculation, the integration of the peaks in quantitative ^{13}C spectra was performed using MestReNova built-in software. Using 80 scans, the threshold concentration in ^{13}C spectrum is around 4 mol%. Combining ^1H NMR spectrum, the threshold concentration is 1 mol%. In the hydrogenated samples, the difference in peak area between C-O and C=O, in both ^1H and ^{13}C spectra, is more than two orders of magnitude. The difference in peak area between C-OH and other oxygen-containing functional groups (hemiacetals, acids, and esters), in the ^{13}C spectrum, is approximately two orders of magnitude.

All Nuclear Magnetic Resonance (NMR) experiments were proceeded on Bruker Avance-500 with a DCH cryoprobe at University of Wisconsin-Madison Chemistry NMR Facility. The sample was dispersed in CDCl_3 (1/7 mass ratio), sonicated, and transferred to an NMR sample tube to obtain ^1H , ^{13}C , and heteronuclear single quantum coherence (HSQC) spectra. For the samples with Co catalyst, the sample was centrifuged before dispersing in CDCl_3 . The parameters for NMR experiments are shown below:

- 1) ^1H NMR (zg30): 32 scans, delay of 1 s, 90-degree pulse width of 9.6 μs
- 2) ^{13}C NMR (zgig): 80 scans, delay of 20 s, 90-degree pulse width of 10.35 μs
- 3) HSQC (hsqcetgp): the acquire size was [1024, 400] with 2 scans per increment, delay of 2s.

Computational Details

All calculations were performed using the *Gaussian 09* software package.(52) High density polyethylene (HDPE) was modeled by an oligomer of $C_{10}H_{22}$ with linear C-backbone, representing the ground-state conformation of HDPE. We used the specific oligomer after checking the effect of chain length on the calculation of transition state (TS) energy; C_{10} represented a structure at which the TS energy was converged, and the computational cost remained reasonable.(53-55) We also checked that rotation of linear $C_{10}H_{22}$ structure to other metastable conformers (cis- or trans-C backbones) does not lead to a decreased TS energy to form paraffinic, mono-olefinic and di-olefinic products. In this study, the preliminary geometric optimizations of all initial molecular structures were performed at the B3LYP level of theory.(56) The resulting structures were then used as initial structures for full optimization using second Møller–Plesset perturbation (MP2) theory.(29) The 6–31++g(d,p) basis set was used for all calculations. (30, 31) All reaction Gibbs free energies and TS energies are reported at the (MP2,6–31++g(d,p)) level of theory at $T=500\text{ }^{\circ}\text{C}$, which is the temperature used in the HDPE pyrolysis experiments.

To determine the TS structure and calculate the respective TS Gibbs free energy, the free energy surface (FES) scan method as implemented in the *Gaussian 09* package was employed at $T=0\text{ K}$ first (MP2 energy obtained at this step).(57) We then added temperature corrections ($T=500\text{ }^{\circ}\text{C}$) to the energetics by vibrational frequency analysis on the structures obtained from the FES scan. The TS was confirmed by only one imaginary frequency corresponding to the reaction coordinate. The activation energy barrier (ΔG^*) was defined as the energy difference between the respective TS and initial state (IS) energy at $T=500\text{ }^{\circ}\text{C}$; the thermodynamic Gibbs free energy of reaction (ΔG) was defined as the energy difference between the respective intermediate/final state (FS) and initial state (IS) at $T=500\text{ }^{\circ}\text{C}$. The TS state for reaction steps involving radicals were determined by the minimum energy crossing point method (see example **Fig. S9** and discussion for details).(58-60)

The thermal depolymerization chemistry of the formation of di-olefins, mono-olefins and paraffins in the pyrolysis oil was investigated using second Møller–Plesset perturbation theory (MP2) (29, 61) and 6-31++G** basis sets (30, 62) as implemented in the *Gaussian 09* package.(52) The activation energy barriers for all radical-mediated reaction steps are estimated by

implementing the minimum free energy crossing point (MECP) (58-60) between free energy profiles with singlet and triplet spin multiplicity.

Supplementary Figures and Text

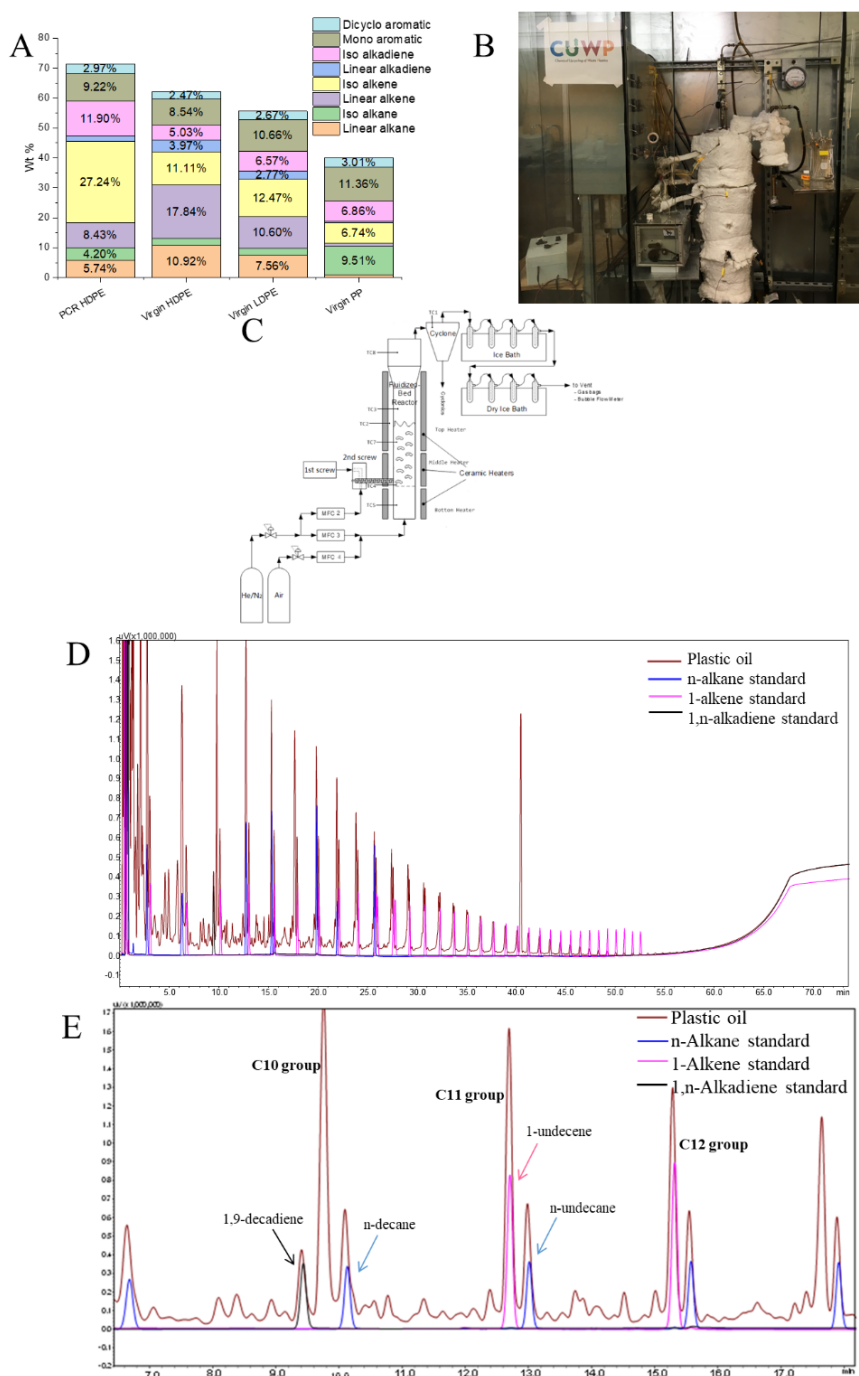


Fig. S2. (A) Product distribution of detailed analysis of non-distilled polyolefin oil with branched information under reaction condition of 500 °C, 20s residence time. This result contains products in the range from C7 to C40. (B) Fluidized bed reactor set-up for plastic pyrolysis. (C) Scheme of the fluidized bed reactor set-up. (D) 1D GC chromatograph of plastic pyrolysis oils and alkane, alkene, and alkadiene standards. (E) Zoomed in 1D GC results in the range of C10 to C12, showing the separation of linear alkane, alkene, and alkadiene.

Pyrolysis of PCR-HDPE generates oil, gas, and char. The yield of oil, gas, and char are 70 wt%, 22.4 wt%, and 2.2 wt%, respectively. To simplify the analysis, the obtained pyrolysis oil was

separated into light cut (<C11, 73.2 wt%) and heavy cut (>C10, 26.8 wt%) by distilling the pyrolysis oil at 165 °C. The obtained oil was characterized by GC, GC×GC-FID, and NMR. The ratio of the linear hydrocarbons was calculated by 1D GC (**Fig. S2E**). The blobs at similar ¹D retention time but smaller ²D retention time refer to the branched alkenes and branched alkadienes (**Fig. 2A Inset**). The carbon number increases with the increased ¹D retention time. The blobs in the middle region of the GC×GC-FID chromatograph belong to mono and multi-cyclic aromatics. The NMR spectrum of the plastic oil is shown in **Fig. 2D** indicating that the plastic pyrolysis oil contains mainly three types of bonds: aliphatic groups, C=C, and aromatics with the absence of the oxygen-containing functional groups (i.e., hydroxyls or aldehydes). The PCR-HDPE pyrolysis oil was also sent to Triton Analytics Corporation for detailed composition analysis by detailed hydrocarbon analysis (DHA) for light cut (**Table S3**) and nitric oxide ionization spectroscopy evaluation (NOISE) for heavy cut (**Table S4**) where both results are consistent with the GC×GC and NMR measurements. The light cut starts from species with 5 carbons with pentene and hexene being the most abundant species. The heavy cut starts to form hydrocarbons with 11 carbons.

Fig. 1 shows that the HDPE has the highest yield of alkenes and alkadienes. Apart from the PCR-HDPE, other polyolefins (virgin HDPE, virgin LDPE, and virgin PP) were also pyrolyzed under the same conditions. The colored PCR-HDPE oil contains less linear products compared to virgin HDPE which likely resulted by the fillers/pigments/additives. We speculate that the impurities containing metals in the PCR-HDPE further lead to the isomerization of the linear hydrocarbons during the pyrolysis process. HDPE generated most alkenes and alkadienes; LDPE produced the pyrolysis oil with a similar product distribution as HDPE; the pyrolysis oil generated from PP shows the least olefins concentration. In general, a straight carbon backbone favors the generation of alkenes and alkadienes. The yield of branched hydrocarbons increases with the increased branch ratio in the feedstock. With the increase of the branch density in the polymers, more paraffins and aromatics were generated. Therefore, under the same pyrolysis condition, the feedstock has fewer branches that could favor the formation of olefin and dienes.

According to **Fig. S3B**, the GC×GC-FID could separate branched hydrocarbons from linear hydrocarbons with carbon numbers higher than C7. Compared with **Fig. S3A**, **Fig. S3B** excluded the C5 and C6 products to better present the branch ratio. The paraffin standard contains C8-C40 saturated alkanes. The concentration for each alkane was 500 µg/mL. We prepared our own 1-alkene standard with octene, nonene, undecane, dodecane, tetradecene, pentadecene, and

heptadecene (C8, C9, C11, C12, C14, C15, and C17). We used 1,9-decadiene (C10) as the alkadiene standard. The concentration of each alkene and alkadiene was 200 $\mu\text{g/mL}$.

In 1D GC-FID (**Fig. S2D-E**), in the same carbon number groups, the linear alkadiene shows the shortest retention time, followed by linear alkene and linear alkane. Therefore, the wt% of the linear n-alkane, 1-alkene, and 1, n-alkadiene for each carbon number can be obtained from the chromatograph based on the standards. This ratio is further used to calculate the percentage of linear alkanes, alkenes, and alkadienes blobs volume of the linear hydrocarbons in the GC $\times$ GC-FID chromatograph of the pyrolysis oil.

Although the 1D GC-FID can separate the linear hydrocarbons, it does not have enough resolution to separate aromatics and dicyclo-aromatics. Because the 1D GC-FID mainly separates compounds through boiling temperatures, when the sample is complicated, peaks could be overlapped. Since GC $\times$ GC-FID can separate in both boiling temperature and polarity, it can reduce the overall overlapping peaks. For the PCR-HDPE pyrolysis oil, with 1D GC-FID, we could see around 250 peaks in the spectrum, while with GC $\times$ GC-FID, we could see more than 1000 blobs in the chromatograph. Therefore, using GC $\times$ GC-FID could provide a more detailed analysis of the plastic oil distribution.

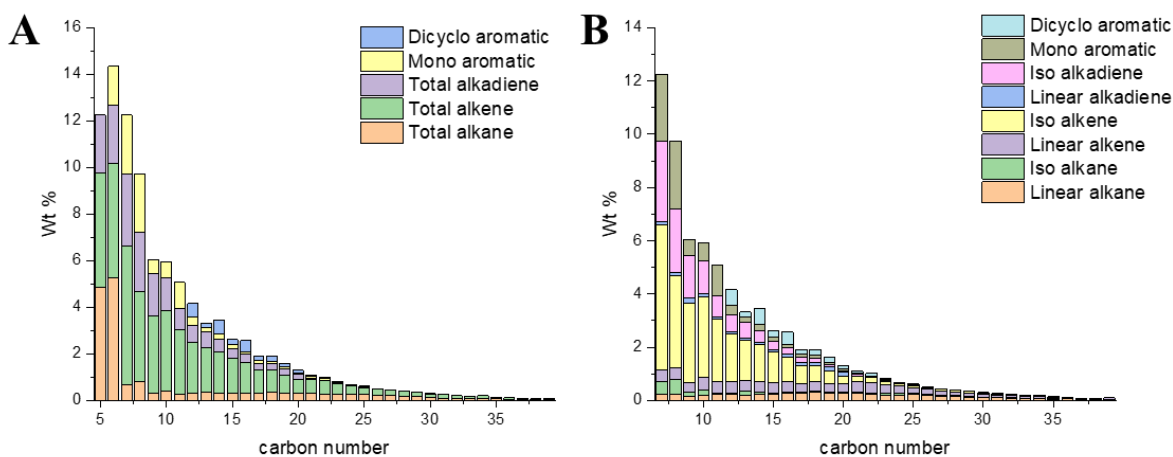


Fig. S3. Product distribution of PCR-HDPE pyrolysis (A) whole oil (C5~C40), and (B) detailed oil analysis with branch information (C7~C40).

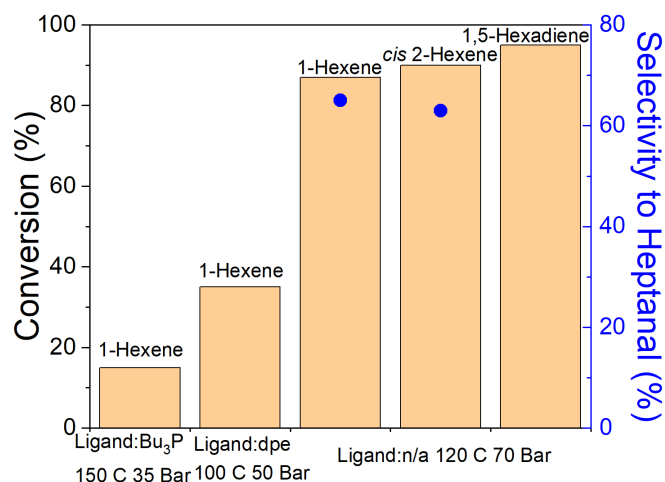


Fig. S4. Hydroformylation of pyrolysis oil model compounds catalyzed by $\text{Co}_2(\text{CO})_8$ under varied conditions. Bu_3P and dpe represent tri-*n*-butylphosphine and 1,2-Bis(diphenylphosphino) ethane monoxide, respectively.

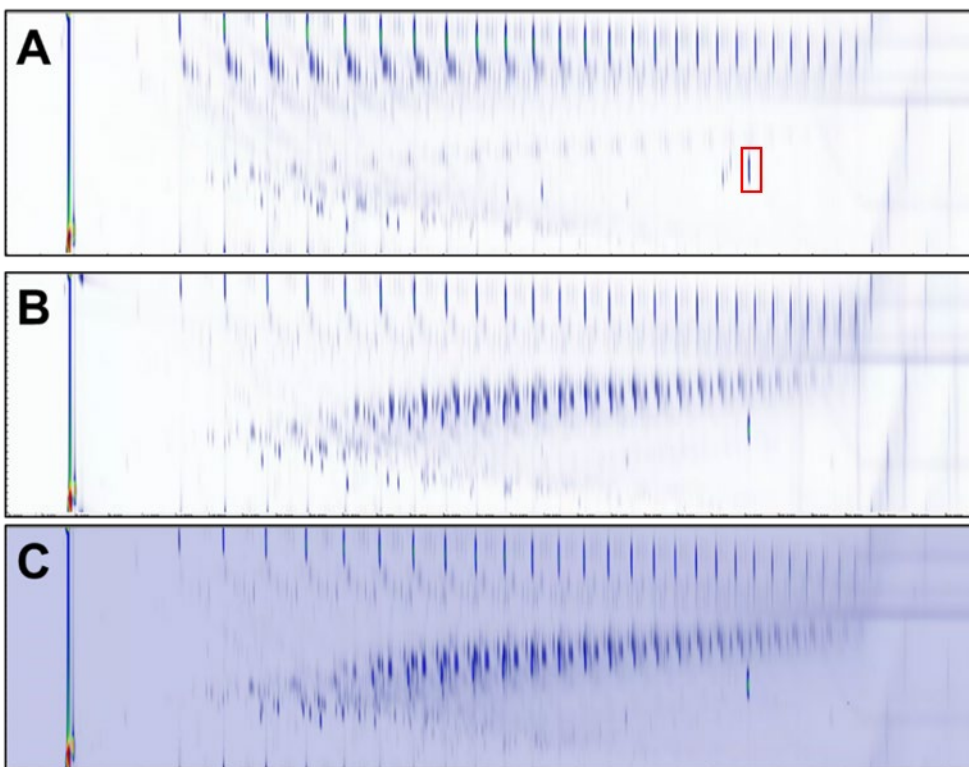


Fig. S5. Hydroformylation of pyrolysis oil (heavy cut) catalyzed by $\text{Co}_2(\text{CO})_8$ and hydrogenation of the “hydroformylated oil” (heavy cut) catalyzed by 20%Ni/SiO₂. The species concentration increases from blue to green color. (A) GCxGC-FID analysis of pyrolysis oil heavy cut. The circled blob was bis(2-ethylhexyl) terephthalate (DEHT), a common plasticizer, predicted by mass spectrometry. (B) GCxGC-FID analysis of the oil (A) after hydroformylation catalyzed by $\text{Co}_2(\text{CO})_8$ in a 350 mL stainless steel batch reactor for 3 h. Conditions: 120 °C, 70 Bar Syngas ($\text{CO}/\text{H}_2=1$), 900 rpm, 10wt% $\text{Co}_2(\text{CO})_8$ in oil. (C) GCxGC-FID analysis of the oil (B after hydrogenation catalyzed by 20%Ni/SiO₂ in a continuous flow reactor for 5 h. Conditions: 100 °C, 78 Bar H_2 , dissolved in (50 wt%) in hexane, weight hourly space velocity (WHSV)=1 h⁻¹.

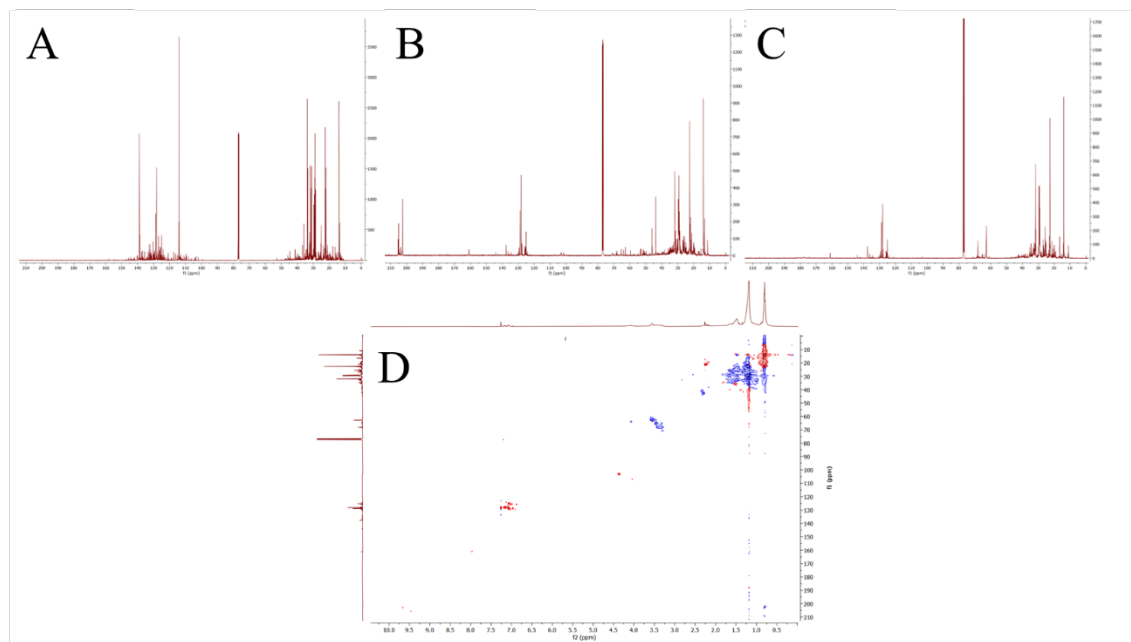


Fig. S6. (A) Full ^{13}C -NMR spectrum of **Fig. 2D** in the main text. (B) Full ^{13}C -NMR spectrum of **Fig. 2E** in the main text. (C) Full ^{13}C -NMR spectrum of **Fig. 2F** in the main text. (D) Full ^1H - ^{13}C HSQC NMR spectrum of **Fig. 2G** in the main text.

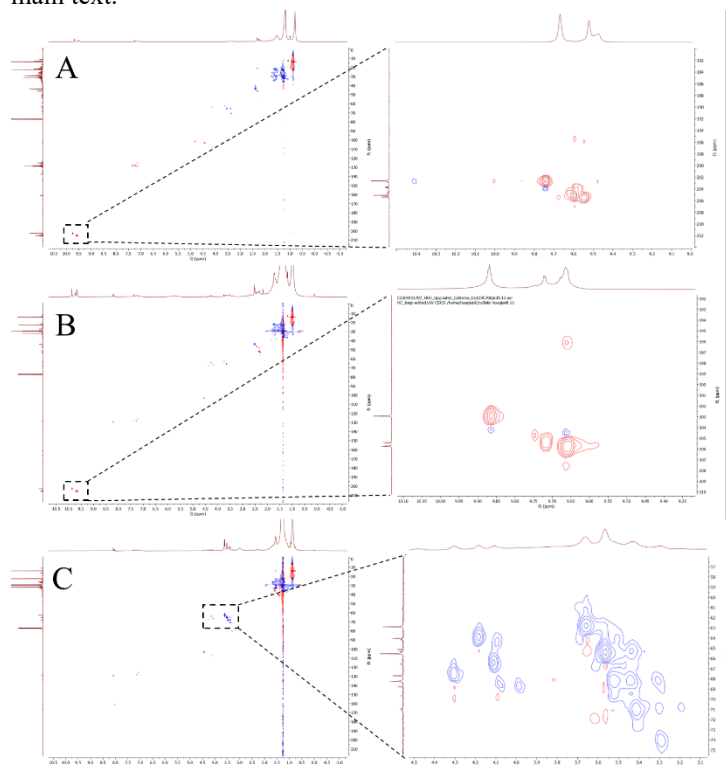


Fig. S7. ^1H - ^{13}C HSQC NMR spectrum of the (A) hydroformylated oil light cut, (B) hydroformylated oil heavy cut, and (C) hydrogenated “hydroformylated oil” heavy cut. Hydrogenation was catalyzed by 20%Ni/SiO₂ in a continuous flow reactor for 5 h. The heavy cut “hydroformylated oil” was diluted in hexane during the reaction. Conditions: 100 °C, 78 Bar H₂, weight hourly space velocity (WHSV)=1 h⁻¹. Blue and red represents primary and secondary carbon, respectively. Spectra on the right represent the zoomed in boxed region.

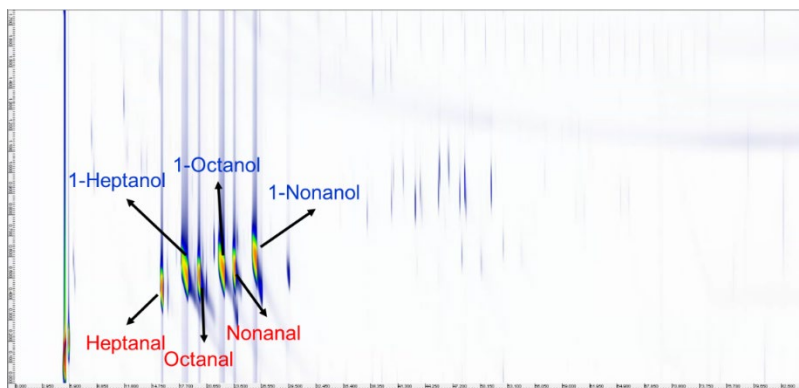


Fig. S8. GC×GC-FID chromatograph of varied standards. Linear alkanol has a larger first-dimensional (¹D) column retention time compared with that of linear alkanol with the same carbon number.

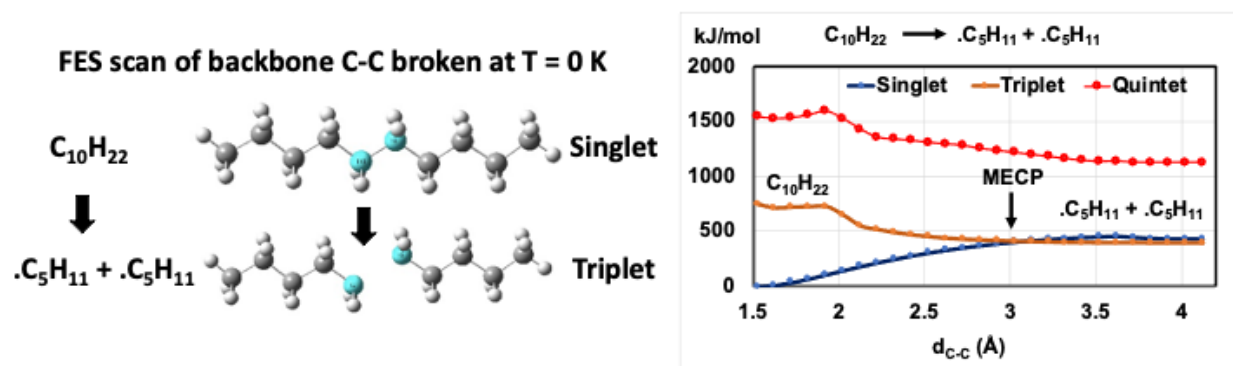


Fig. S9. For reaction steps involving radicals, the minimum energy crossing point (MECP) method was employed to determine the transition state (TS) and its energy (spin-multiplicity changes along the bond-breaking reaction coordinate).

Fig. S9 shows an example of implementing the minimum energy crossing point (MECP) method for determining the TS of reactions involving radicals in the HDPE pyrolysis reaction mechanism. In particular, **Fig. S9** shows the energetics along the C-C bond breaking reaction coordinate in $C_{10}H_{22}$ oligomer to form two $.C_5H_{11}$ radicals. All radical-mediated reactions discussed in this contribution were studied using this MECP analysis. During the Free Energy Scan (FES) of the selected C-C bond to be broken (**Fig. S9**, blue line, distance increases with a step of 0.1 Å), two identical C_5H_{11} radicals were formed (no H abstraction in this step). The energies of $C_{10}H_{22}$ and the $.C_5H_{11}$ radical are calculated in their singlet, triplet, and quintet multiplicity to determine their ground-state. Accordingly, two $.C_5H_{11}$ radicals in the triplet spin state (one unpaired electron in each radical) are confirmed as the ground-state, whereas the initial state (IS) of the linear $C_{10}H_{22}$ oligomer is in the singlet electronic configuration. For such a reaction with spin change between the IS and the final state (FS), the spin-state surface crossing is determined

by scanning the FES with singlet, triplet, and quintet multiplicity. The structure at minimum energy crossing point (MECP) between the FES of singlet and triplet determines the transition state (TS) structure, which was then used for vibrational calculations to determine the TS free energy at 500 °C. The MECP method has been widely used to estimate the TS energy for this type of reactions.^(59, 60, 63-65)

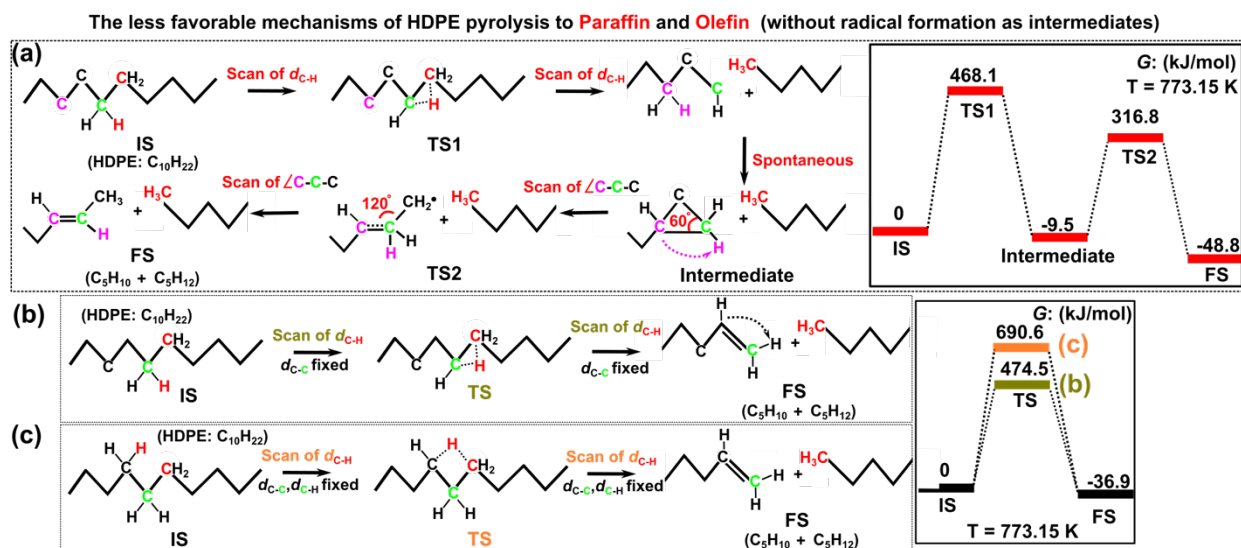


Fig. S10. Thermodynamically and kinetically less favorable mechanisms (compared with that shown in **Fig. 4A**) for the formation of paraffin, mono-olefin, and di-olefin. Linear decane ($C_{10}H_{22}$) was used as a surrogate molecule for HDPE and yielded pentane (C_5H_{12}) and pentene (C_5H_{10}) through pyrolysis, without involving radicals in these mechanisms.

Different from the radical-mediated mechanism (in **Fig. 4A**) to form pentane (C_5H_{12}) and pentene (C_5H_{10}) by scanning (increasing) the bond distance of backbone C-C in linear decane ($C_{10}H_{22}$), we also investigated other mechanisms starting with H abstraction by scanning the bond distance between C and adjacent H (in **Fig. S10 (a) and (b)**) or between C and next adjacent H (**Fig. S10 (c)**). All three mechanisms presented in **Fig. S10** are radical-free, and their TS energies were determined to be higher than that (352.9 kJ/mol) for the radical-mediated mechanism (shown in **Fig. 4A**). Specifically, the mechanism in **Fig. S10 (a)** starts from hydrogen (H highlighted in red) transfer by decreasing the distance between the selected C and H atoms (both highlighted in red) for an FES scan. As a result, the pentane (C_5H_{12}) and a metastable C_5H_{10} isomer with a C-triangular ring are formed as **Intermediate** with a TS energy (ΔG^*) of 468.1 kJ/mol. Then, 326.3 kJ/mol energy is needed to transform the specific triangular intermediate C_5H_{10} to its ground-state internal olefin (C_5H_{10}). In **Fig. S10 (b)**, during the FES scan in which the distance between the two selected C and H atoms is decreasing (atoms highlighted in red, **Fig. S10 (b)**), we also fix the bond

distance of backbone C-C (C atoms highlighted in green and black of the **IS** structure) to avoid the formation of the triangular metastable C_5H_{10} structure. With this additional constraint, a mechanism to form the terminal olefin (C_5H_{10}), which is not the most stable isomer, is obtained with a TS energy ($\Delta G^* = 474.5$ kJ/mol), which is higher than that ($\Delta G^* = 468.1$ kJ/mol, in **Fig. S10 (a)**) of H transfer without any backbone C-C constraints. Finally, **Fig. S10 (c)** shows yet another mechanism we evaluated its energetics: forming terminal olefin C_5H_{10} whereby C abstracts a H from two carbons apart. This mechanism is the most energetically demanding ($\Delta G^* = 690.6$ kJ/mol) among all those we studied, and therefore is expected to play no role in the pyrolysis reaction.

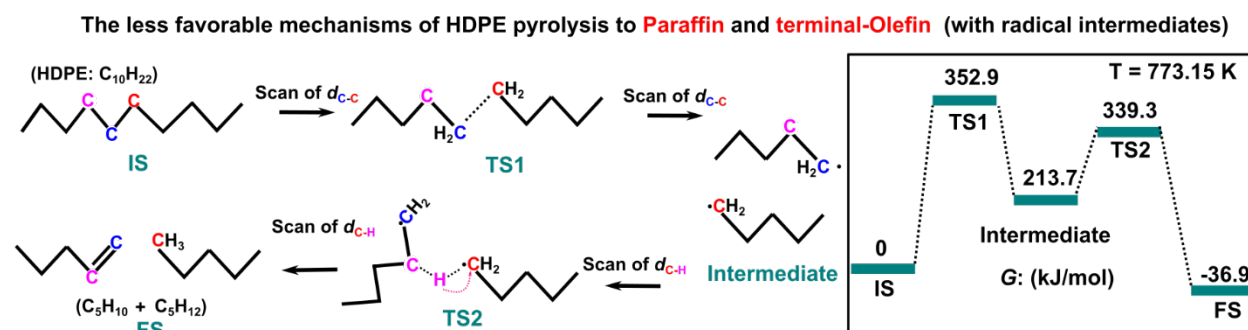


Fig. S11. A radical-mediated mechanism (which is thermodynamically and kinetically less favorable than the mechanism shown in **Fig. 4A**) for linear decane ($C_{10}H_{22}$) to pentane (C_5H_{12}) and terminal pentene (C_5H_{10}) reaction.

Fig. S11 shows another less favorable radical-mediated mechanism (compared to the mechanism shown in **Fig. 4A**) for linear decane ($C_{10}H_{22}$) pyrolysis to paraffins and mono-olefins. The steps from **IS** to **Intermediate** are the same as shown in **Fig. 4A** by breaking a backbone C-C bond to form two $\cdot C_5H_{11}$ radicals. Starting from **Intermediate**, one H originally bonded to the secondary terminal C site of one of the two $\cdot C_5H_{11}$ radicals (H atom highlighted in pink, see **TS2** of **Fig. S11**) is transferred to saturating the other $\cdot C_5H_{11}$ radical. Therefore, the ultimate products of paraffin (C_5H_{12}) and terminal olefin (C_5H_{10}) are formed, however, the barrier ($\Delta G^* = 125.6$ kJ/mol) of such a H abstraction step (**Intermediate** $\rightarrow$ **FS**) is higher than the respective one ($\Delta G^* = 119.4$ kJ/mol) in **Fig. 4A**, where an internal olefin (C_5H_{10}) was formed.

In conclusion, we have provided five different reaction mechanisms for HDPE (surrogate molecule: $C_{10}H_{22}$) pyrolysis to paraffin (C_5H_{12}) and mono-olefin (C_5H_{10}); see **Fig. 4A**, **Fig. S10**, and **Fig. S11**. The radical mechanism for forming paraffin (C_5H_{12}) and internal olefin (C_5H_{10}) proposed in **Fig. 4A** is the energetically preferred mechanism with a TS energy of 352.9 kJ/mol and reaction energy of -48.8 kJ/mol (see a summary comparison in **Table S6**).

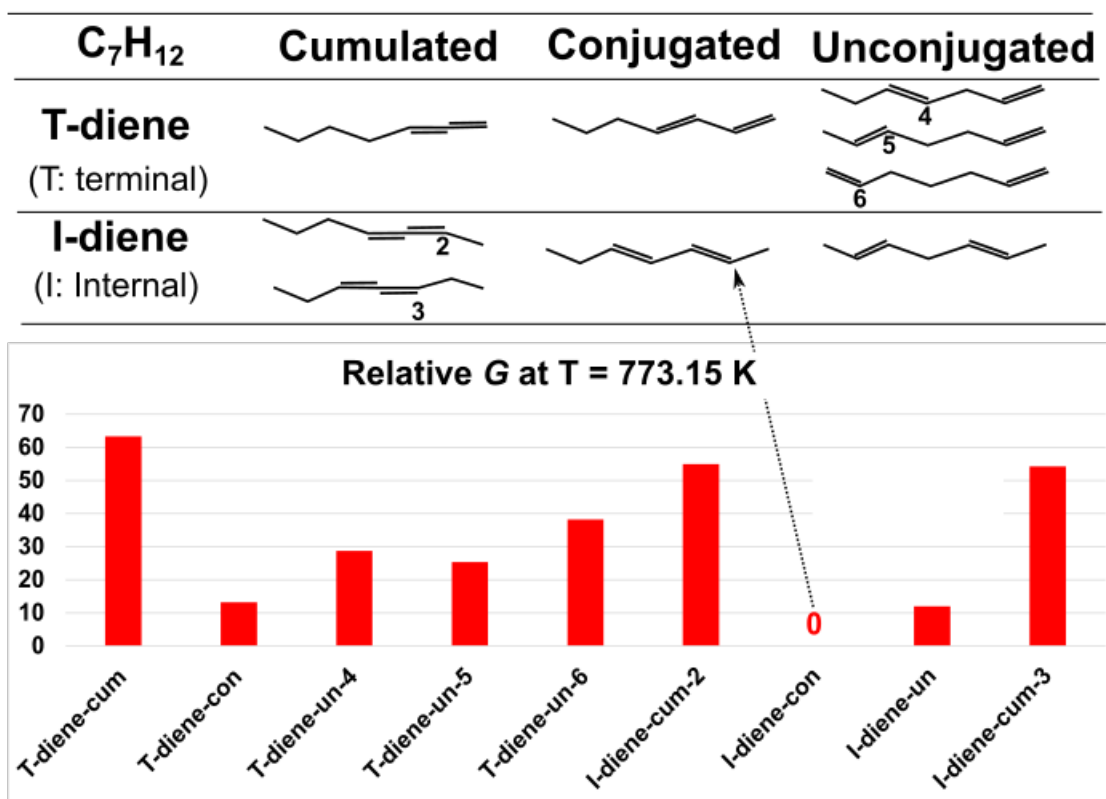


Fig. S12. Relative thermodynamic stability of linear Heptadiene (C_7H_{12}) isomers at $T=773.15K$, where the Gibbs free energy of the internal conjugated C_7H_{12} (ground-state) is set to 0 kJ/mol.

Thermodynamics of HDPE pyrolysis pathways (P1, P2, P3) to form Diene (C₇H₁₂)

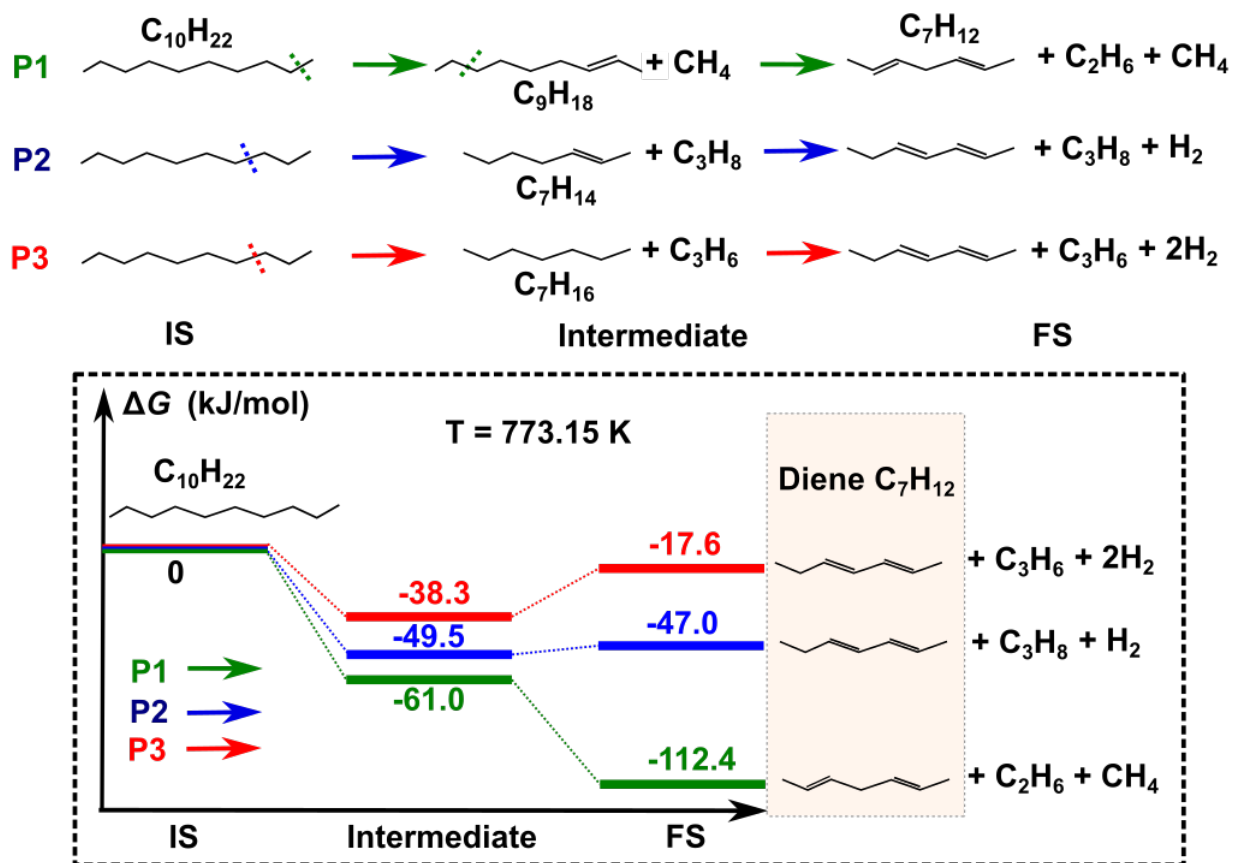


Fig. S13. Thermochemical analysis of three possible pathways for linear decane (C₁₀H₂₂) pyrolysis to heptadiene (C₇H₁₂).

Fig. S12 shows the relative thermochemical stability of all isomers of linear heptadiene (C₇H₁₂) at T=773.15 K. The conjugated isomers are more stable than the cumulated and unconjugated in both terminal and internal heptadiene. Among all nine isomers, the internal conjugated heptadiene (C₇H₁₂) is confirmed as the ground-state structure, which is selected as the representative heptadiene products of pathways (P2 and P3) in **Fig. S13** for Gibbs free energy estimates of the overall reactions, whereas pathway (P1) prefers to form the internal unconjugated heptadiene (C₇H₁₂) as the preliminary pyrolysis product (see details in **Fig. 4B** for the respective mechanism). According to the thermochemical comparison of reaction Gibbs free energies of P1-P3 (**Fig. S13**), P1 is the most exergonic path with backbone C-C scission from both end sides of the C₁₀H₂₂ molecule, leading to simultaneous production of C₂H₆ and CH₄, in addition to C₇H₁₂. Moreover, P1 is also found to be the kinetically most favorable path with the lowest TS energy for C-C bond scission and H abstraction steps (see details in **Fig. 4B**). In comparison, both P2 and P3

require one or two steps of dehydrogenation after the backbone C-C scission, while the dehydrogenation process is endergonic ($\Delta G > 0$) at $T=773.15$ K. In addition, after a full mechanistic study of **P2**, the second most likely path per **Fig. S13**, we found the dehydrogenation step is characterized by a much higher TS energy ($\Delta G^* = 564.2$ kJ/mol) (see **Fig. S14**), compared to that ($\Delta G^* = 350 \sim 400$ kJ/mol) of backbone C-C dissociation. Therefore, we conclude that **P1**, which does not involve H_2 evolution, dominates the path to form heptadiene (C_7H_{12}), a result that is in agreement with our experimental findings.

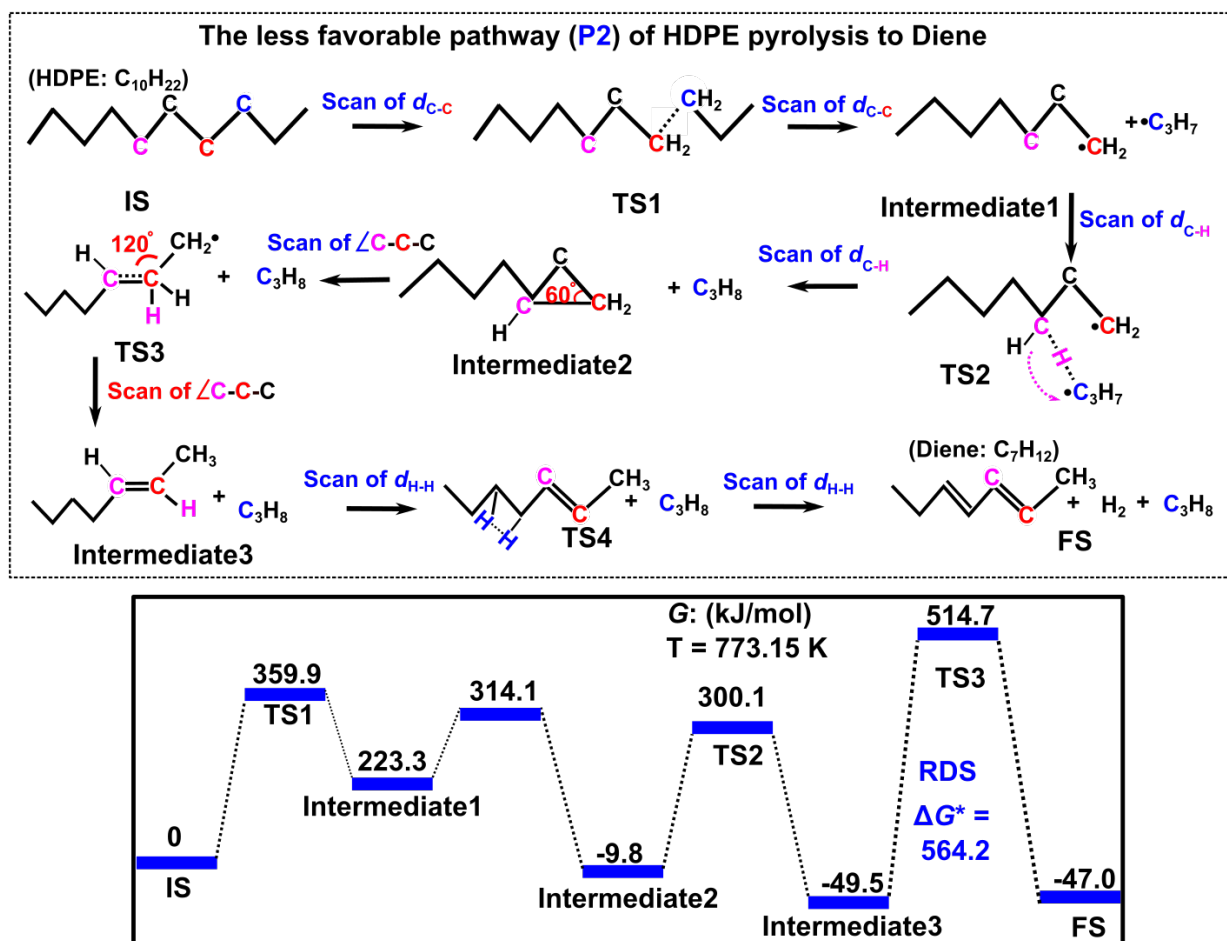


Fig. S14. The mechanism and Gibbs free energy profile for P2 (in **Fig. S13**): linear decane ($C_{10}H_{22}$) pyrolysis to heptadiene (C_7H_{12}). To be compared with the most favorable pathway (P1) is shown in **Fig. 4B**.

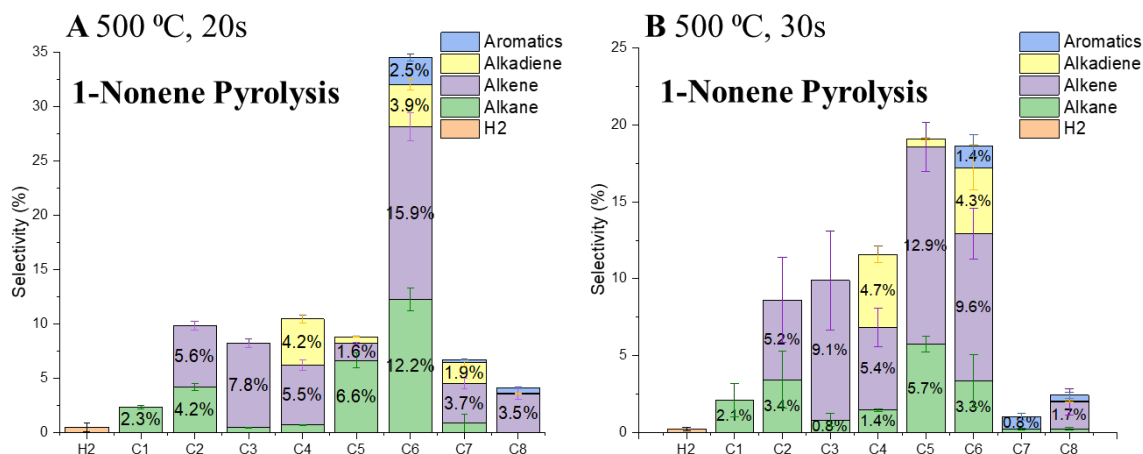


Fig. S15. 1-nonene pyrolysis product distribution at 500 °C at various residence times. (A) 20 s with a conversion of 10.5% and (B) 30 s with a conversion of 28.0%. At least 3 samples were collected in each experimental condition.

Fig. S15 shows 1-nonene pyrolysis (partial pressure 16.2 kPa) under 500 °C with residence time 20 s and 30 s, respectively. We had to use 1-nonene in our experiments, as 2-nonene used in our theoretical calculation was not commercially available. The pyrolysis mechanism for the two isomers is likely similar. According to our mechanism in **Fig. 4B**, the backbone C-C breaking in nonene (**Int3**) to yield heptadiene (**FS**) should start from the other terminal side, which is away from the characteristic C=C bond in both 1-nonene (experimental sample) and 2-nonene (**Int3**, in **Fig. 4B**). **Fig. S15A** shows the formation of heptadiene (yellow bar in C7 column) and ethane byproduct (green bar in C2 column), which is performed under the same condition (500 °C, 20 s) as we reported for the experimental pyrolysis of HDPE. Some of the heptadiene formed is decomposed to propylene and butadiene. This can explain why a higher heptadiene selectivity (1.9%) versus ethane selectivity (4.2%) is observed at 500 °C, 20 s (**Fig. S15A**). To confirm this, we also performed pyrolysis at 30 s residence time. As shown in **Fig. S15B**, the produced heptadiene (at 20 s) was completely decomposed with an increase in propylene and butadiene. In addition, our theoretical mechanistic work suggested that the other paths for heptadiene formation with H₂ as the byproducts (**Fig. S13-14**) are much less favorable compared to the path producing ethane byproduct (**Fig. 4B**). This prediction is in good agreement with our experimental findings that a very small amount of H₂ is produced during 1-nonene pyrolysis (**Fig. S15**).

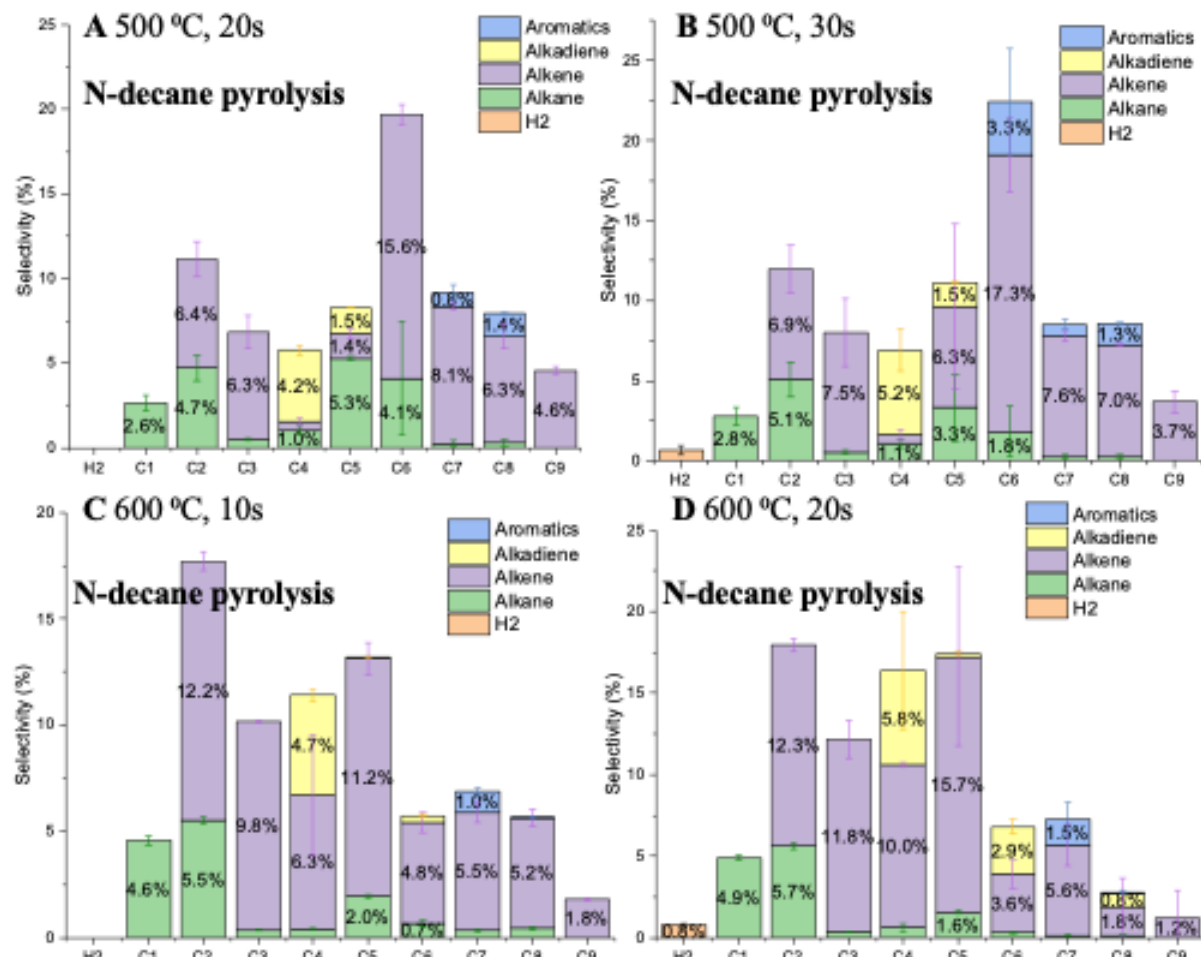


Fig. S16. N-decane pyrolysis product distribution at two temperatures and residence times. (A) 500 °C, 20 s with a conversion of 5.7%, (B) 500 °C, 30 s with a conversion of 6.7%, (C) 600 °C, 10 s with a conversion of 18.2%, and (D) 600 °C, 20 s with a conversion of 29.6%.

We also performed experiments to compare against our theoretically-derived mechanism for HDPE decomposition to alkanes and alkenes. In our theoretical work we modeled HDPE with N-decane. **Fig. 4A** shows the mechanism to produce pentane and pentene. Our experimental pyrolysis of N-decane yielded very limited conversion at 500 °C (5.7–6.7%, in **Fig. S16A-B**), which was the temperature we performed HDPE pyrolysis in the main manuscript. We increased the N-decane pyrolysis temperature to 600 °C and the conversion (**Fig. S16C-D**) was almost quadrupled compared to that at 500 °C. At 600 °C, we observed a normal distribution of products centered around C₅ hydrocarbons (supporting our mechanism in **Fig. 4A**). The alkanes are lower in concentration than the alkenes, especially for C₃, which can be contributed by the secondary decomposition of C₃–C₉ alkanes at this higher temperature. Accordingly, a high yield of C₂ and C₁ products was observed at 600 °C. We observe a normal product distribution with C₅ being the highest C₃₊ product, which is in good agreement with our theoretical predictions for the main products (alkanes and alkenes).

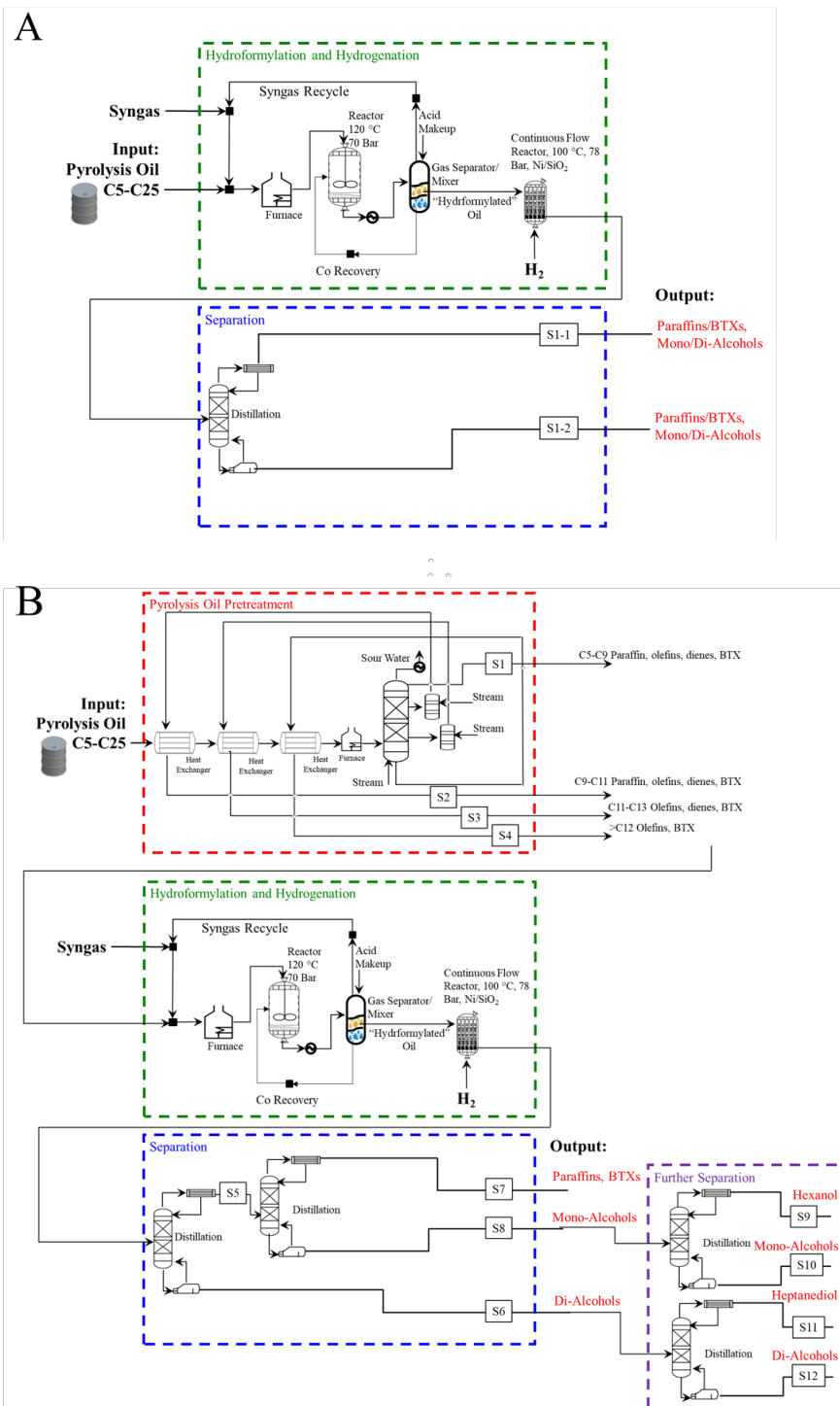


Fig. S17. Proposed process flow diagram to produce paraffins, BTXs, mono-alcohols, and di-alcohols from waste plastic pyrolysis oil via hydroformylation and hydrogenation. (A) Process without pyrolysis oil pretreatment; (B) process including the pyrolysis oil pretreatment into different cuts followed by hydroformylation, hydrogenation and separation of each cut.

A mixture of paraffins, BTXs, mono-alcohols and di-alcohols are obtained (**Table S8**) via the route of hydroformylation, hydrogenation, and separation (**Fig. S17A**). The separation of di-alcohols and mono-alcohols is challenging considering their similar boiling point (e.g., C8 di-alcohol 279 °C vs. C13 mono-alcohol 280 °C, **Table S9**). Instead, fractional distillation of the pyrolysis oil into stream S1, S2, S3, and S4 (**Fig. S17B**) followed by hydroformylation of each stream. PetroFac Column (Aspen Plus V12.1) was used to simulate the separation of plastic pyrolysis oil into 4 streams followed by hydroformylation, hydrogenation, and separation of each stream. Species with >0.6 wt% (**Table S3** and **S4**) in the pyrolysis oil were used for simulation and normalized to total flow rate at 100 kg/h. The pretreatment of the pyrolysis oil separates their precursors before hydroformylation and hydrogenation, thus enabling the separation of more valuable di-alcohols. The difference in boiling point between target products in each cut (**Table S9**) enables the separation of more valuable mono-alcohols and di-alcohols. As depicted in **Table S10**, high purity (S6 in **Fig. S17B**, 500 ppm impurities) di-alcohol and mono-alcohol (S8 in **Fig. S17B**, 3000 ppm paraffins and BTXs) products can be obtained via this route. Further purification of the stream enables the separation of individual mono-alcohols or di-alcohols with high-purity. For example, 91% hexanol (S9 in **Fig. S17B**) and 99.5% heptanediol (S11 in **Fig. S17B**) can be obtained as described in **Table 10**.

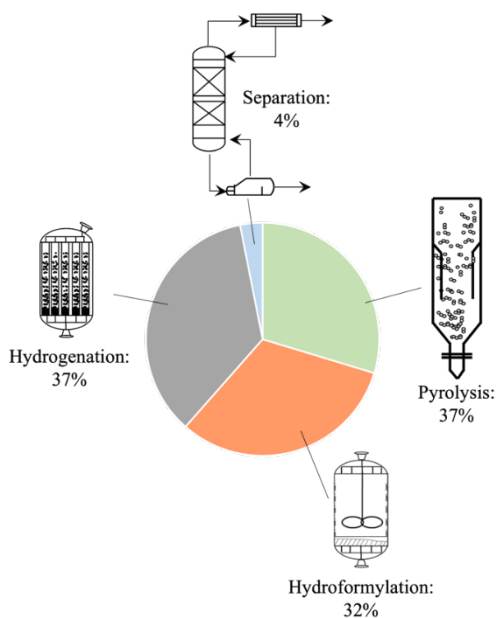


Fig. S18. Breakdowns of carbon emissions.

Fig. S18 shows that carbon emissions from pyrolysis, hydroformylation, and hydrogenation each account for approximately one-third, while emissions from separation account for 4%.

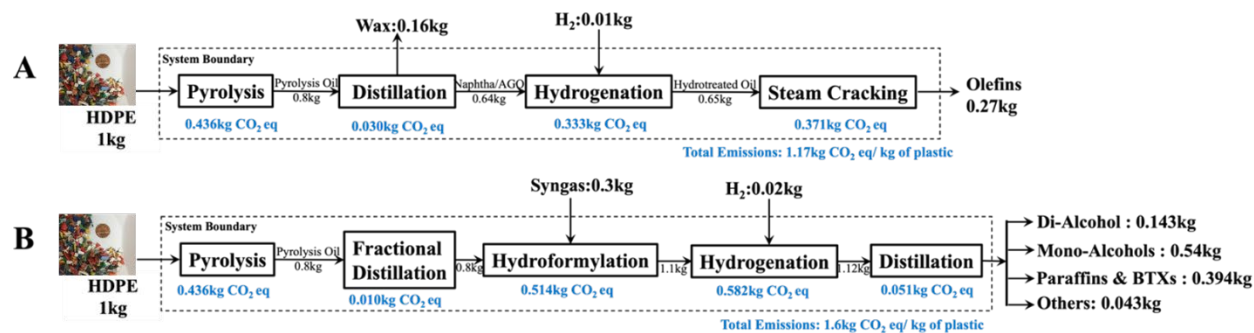


Fig. S19. System boundaries of environmental assessments and carbon emissions of individual steps. (A) plastic-to-olefins route. (B) hydroformylation route.

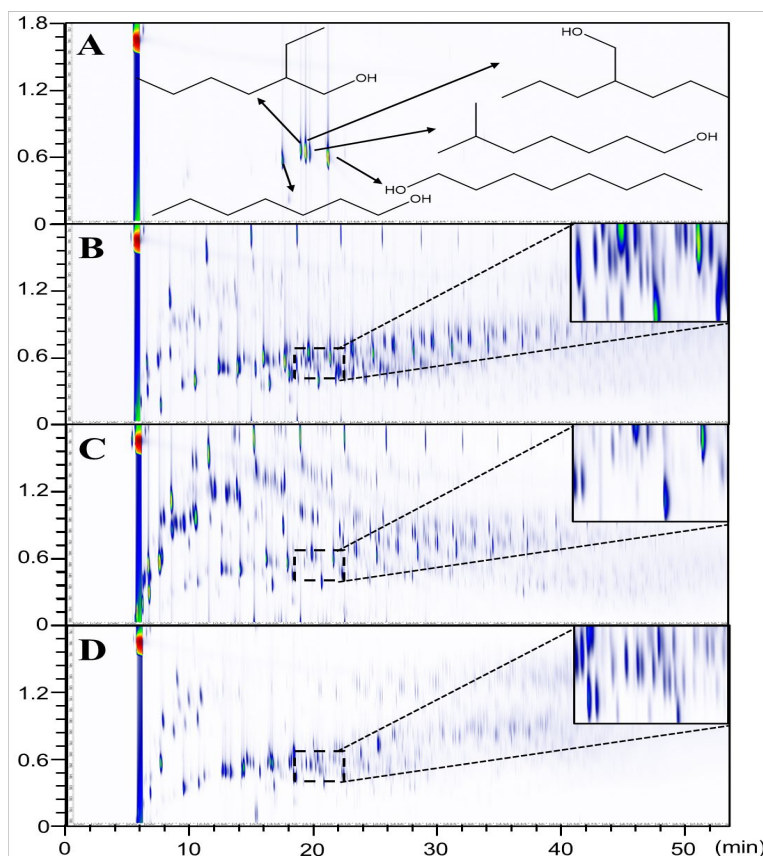


Fig. S20 GCxGC analysis of (A) 1-heptanol, 1-octanol, and branched C8 alcohols; (B) hydroformylated virgin HDPE oil after hydrogenation; (C) hydroformylated virgin LDPE oil after hydrogenation; (D) hydroformylated virgin PP oil after hydrogenation. Hydrogenation was catalyzed by 5wt%Ru/C in a batch reactor for 15 hours (100 °C, 70 Bar H₂, 14wt% catalyst in the hydroformylated oil). Insets in (B-D) represent the zoom-in boxed region.

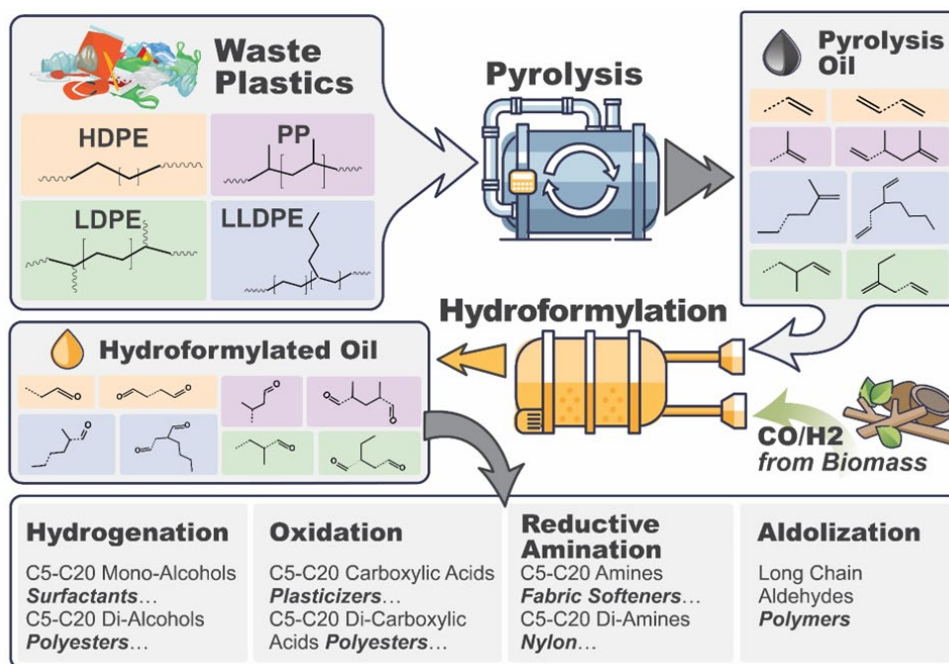


Fig. S21 Proposed routes to produce alcohols, carboxylic acids, and amines from waste polyolefins.

As shown in **Fig. S20**, GC×GC was used to compare the branching properties of the obtained alcohols from different plastics. 2-propanol (sample/2-propanol=1/10 mass ratio) was used as solvent in GC×GC analysis for better dissolvability of the generated alcohols. Based on GC×GC analysis of the standards (**Fig. S20A**), a region of 18.25-22.25 min ¹D retention time and 0.45-0.65 s ²D retention time (boxed region) belonging to C8 alcohols in different upgraded oils were analyzed to simplify the comparison. The position and the volume of the blobs were obtained by the built-in software. A drastically higher ratio of 1-octanol to branched C8 alcohols is shown for HDPE and LDPE compared with PP pyrolysis oil (**Table S14**) suggesting varied branching properties in the generated alcohols. **Fig. S21** describes the proposed routes producing value-added chemicals with varied structures from waste polyolefins.

Supplementary Tables

Table S1. Product distribution of different plastic pyrolysis oils.

Plastics	Gas (wt%)	Liquid (wt%)	Solid(wt%)
Colored PCR-HDPE	22.4	69.8	2.2
HDPE	24.9	72.5	0.9
LDPE	26.3	68.3	1.1
PP	18.8	74.1	1.6

Table S2. Hydroformylation of hexene, 1,5-hexadiene, and pyrolysis oil catalyzed by $\text{Co}_2(\text{CO})_8$. Toluene was used as a solvent for entries 1-5 while solvent was absent for entries 6-7 (see details in Methods).

Entry	Substrate	Temperature (C°)	Ligand	Aldehydes Yield (%)	Pressure (Bar)
1	1-hexene	120	n/a	87	70
2	1-hexene	150	Bu_3P	15	35
3	1-hexene	100	dpe	35	50
4	<i>cis</i> -2-hexene	120	n/a	90	70
5	1,5-hexadiene	120	n/a	95	70
6	Pyrolysis oil light cut	120	n/a	>90	70
7	Pyrolysis oil heavy cut	120	n/a	>75	70

Table S3. Detailed hydrocarbon analysis (DHA) of the pyrolysis oil light cut.

Carbon Number	Paraffins (wt %)	N-Paraffins (wt %)	Olefins (wt %)	Naphthenes (wt %)	Aromatics (wt %)	Unknown (wt %)	Total (wt %)
1	0.0044	0.0000	0.0000	0.0000	0.0000	0.0000	0.0044
2	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
3	0.0000	0.0000	0.0014	0.0000	0.0000	0.0000	0.0014
4	0.0092	0.0000	0.1334	0.0000	0.0000	0.0000	0.1426
5	0.2235	0.0040	2.0392	0.0327	0.0000	0.0000	2.2994
6	0.5274	0.6923	11.9474	0.3077	2.8218	0.0630	16.3596
7	1.4683	0.5815	14.5501	0.5754	4.6821	0.7673	22.6248
8	1.4693	3.3144	9.1851	2.2099	4.5116	1.0956	21.7860
9	1.0992	2.3847	6.3879	3.0795	2.0715	1.2211	16.2439
10	0.8485	1.6556	5.3578	0.5383	2.8259	0.4424	11.6659
11	0.5245	1.0822	2.3363	0.0500	0.59650	0.7606	5.3501
12	0.2423	0.3105	0.9287	0.0353	0.2600	0.2725	2.0493
13	0.1349	0.0049	0.3718	0.0000	0.0000	0.2719	0.7835
total	6.5489	10.301	53.2391	6.8290	17.7694	4.8945	94.4164

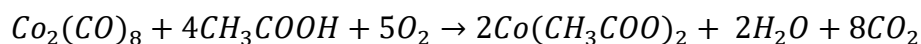
Table S4. Nitric oxide ionization spectroscopy evaluation (NOISE) for the pyrolysis oil heavy cut.

Carbon Number	Paraffins (wt%)	N-Paraffins (wt%)	Olefins (wt%)	Benzenes (wt%)	Dicyclo-benzenes (wt%)
9				0.03	0.05
10	0.08	0.07	0.33	0.17	0.24
11	0.54	0.43	2.37	0.30	0.26
12	0.99	0.83	4.13	0.33	0.16
13	1.31	1.09	5.25	0.24	0.11
14	1.61	1.32	6.60	0.18	0.09
15	1.86	1.56	6.64	0.15	0.07
16	2.00	1.67	6.32	0.12	0.05
17	2.12	1.76	5.95	0.09	0.04
18	2.22	1.86	5.51	0.08	0.03
19	2.27	1.88	4.71	0.07	0.03
20	2.15	1.79	4.06	0.06	0.03
21	2.05	1.70	3.22	0.06	0.02
22	1.85	1.55	2.66	0.05	0.02
23	1.75	1.48	1.98	0.04	0.02
24	1.43	1.23	1.54	0.03	0.02
25	1.22	1.00	1.14	0.03	
26	0.97	0.83	0.85	0.03	
27	0.74	0.62	0.60	0.02	
28	0.59	0.49	0.46	0.01	
29	0.42	0.35	0.36	0.02	
30	0.35	0.28	0.37	0.01	
31	0.29	0.23	0.30		
32	0.24	0.19	0.25		
33	0.19	0.14	0.20		
34	0.16	0.11	0.16		
35	0.14	0.10	0.13		
36	0.16	0.08	0.19		
37	0.11	0.06	0.11		
38	0.10	0.05	0.07		
39	0.09	0.04	0.07		
40	0.07	0.02	0.04		

Table S5. Hydrogenation of hydroformylated pyrolysis oil. Reaction was carried out at 100 °C, 78 bar H₂. Selectivity to alcohols is above 85% for all the data. The substrate to solvent weight ratio was adjusted to achieve a similar weight ratio in the pyrolysis oil light cut. ^a: a continuous flow reactor was used for the hydrogenation step, weight hourly space velocity (WHSV)=1 h⁻¹.

Entry	Catalyst	Feed	Solvent	Reaction Time (h)	Alcohols Yield (%)
1	5%Ru/C	Octanal	Hexane	6	100
2	5%Ru/C	Octanal with 10 wt% Co ₂ (CO) ₈	Hexane	24	5
3	5%Ru/C	Hydroformylated 1,5 hexanediene	Octane	24	90
4	5%Ru/C	Hydroformylated 1,5 hexanediene with 10 wt% Co ₂ (CO) ₈	Octane	24	5
5	5%Ru/C	Hydroformylated pyrolysis oil light cut with 10 wt% Co ₂ (CO) ₈	n/a	12	35
6	5%Ru/C	Hydroformylated pyrolysis oil light cut	n/a	6	89
7	20%Ni/SiO ₂	Hydroformylated pyrolysis oil light cut	n/a	6	80
8 ^a	20%Ni/SiO ₂	Hydroformylated pyrolysis oil light cut	n/a	5	97
9	5%Ru/C	Hydroformylated pyrolysis oil light cut with 10 wt% Co ₂ (CO) ₈	Hexane	12	35
10	5%Ru/C	Hydroformylated pyrolysis oil light cut with 10 wt% Co ₂ (CO) ₈	n/a	24	40
11	5%Ru/C	Hydroformylated pyrolysis oil light cut with 10 wt% Co ₂ (CO) ₈	n/a	36	50
12	10%Cu/ZrO ₂	Hydroformylated pyrolysis oil light cut with 10 wt% Co ₂ (CO) ₈	n/a	12	37
13	20%Ni/SiO ₂	Hydroformylated pyrolysis oil light cut with 10 wt% Co ₂ (CO) ₈	n/a	12	50
14	5%Ru/C	Hydroformylated pyrolysis oil heavy cut	n/a	6	90
15 ^a	20%Ni/SiO ₂	Hydroformylated pyrolysis oil heavy cut	Hexane	5	91

To verify the effects of Co₂(CO)₈, hydrogenation of the octanal in the presence of Co₂(CO)₈ was performed. Drastic decrease in the conversion at longer reaction time revealed that Co₂(CO)₈ inhibits hydrogenation (**Table S5 Entry 2**).



Co₂(CO)₈ was removed via oxidation as depicted above. After the removal of Co₂(CO)₈, the conversion of 1,5-hexadiene hydroformylation products increased drastically from 5% to 95% (**Table S5 Entries 3-4**). Therefore, the Co catalyst in the “hydroformylated pyrolysis oil” was removed before carrying out the hydrogenation step.

Table. S6. A summary of the highest transition state (TS) energy (G^{TS} , kJ/mol) and total reaction energies (ΔG^* , kJ/mol) for five different mechanisms studied for linear decane ($\text{C}_{10}\text{H}_{22}$) pyrolysis to pentane (C_5H_{12}) and pentene (C_5H_{10}); see **Fig. 4A**, **Fig. S10**, and **Fig. S11**. Temperature: 773.15K.

Mechanisms	Fig. 4A	Fig. S10(a)	Fig. S10(b)	Fig. S10(c)	Fig. S11
Highest TS energy (G^{TS})	352.9	468.1	474.5	690.6	352.9
Total reaction energy (ΔG)	-48.8	-48.8	-36.9	-36.9	-36.9

Table S7. Comparison of di-olefin and hydrogen yeild in different polyethylene pyrolysis at 500 °C, 20s residence time

	Di-Olefin (wt%)	H ₂ (wt%)
PCR HDPE	13.5 %	0.5 %
Virgin HDPE	9.0 %	0.9 %
Virgin LDPE	9.3 %	0.8 %

Table S8. Stream data for the separation of final products (**Fig. S17A, Separation**) simulation using Aspen Plus V12.1. Di-alcohols are highlighted in blue. ARO represents monoaromatic species and DIARO represents diaromatic species.

Pyrolysis Oil	Feed	Corresponding Final Products (kg/h)	Corresponding Final Products S1-1 (kg/h)	S1-2 (kg/h)	S1-3 (kg/h)
Sum	100.00	127.42		95.88	31.54
C5H12	5.97	5.97	Pentane	5.97	0.00
C5H10	6.09	9.02	Hexanol	9.02	0.00
C5H8	3.11	6.03	Heptanediol	6.03	0.00
C6H14	6.59	6.59	Hexane	6.59	0.00
C6H12	6.09	8.41	Heptanol	8.41	0.00
C6H10	3.86	6.86	Octanediol	6.86	0.00
C7H16	0.87	0.87	Heptane	0.87	0.00
C7H14	7.34	9.76	Octanol	9.76	0.00
C7H12	3.86	6.44	Nonanediol	3.76	2.68
C8H18	1.00	1.00	Octane	1.00	0.00
C8H16	4.85	6.26	Nonanol	6.26	0.00

C8H14	3.11	4.91	Decanediol	0.03	4.88
C9H20	0.37	0.37	Nonane	0.37	0.00
C9H18	4.10	5.13	Decanol	5.13	0.00
C9H16	2.24	3.40	Undecanediol	0.00	3.40
C10H22	0.50	0.50	Decane	0.50	0.00
C10H20	4.35	5.35	Undecanol	5.35	0.00
C10H18	1.74	2.54	Dodecanediol	0.00	2.54
C11H22	3.48	4.21	Dodecanol	4.19	0.03
C11H20	1.12	1.59	Tridecanediol	0.00	1.59
C12H24	2.74	3.26	Tridecanol	2.23	1.02
C12H22	0.87	1.20	Tetradecanediol	0.00	1.20
C13H26	2.36	2.79	Tetradecanol	0.21	2.58
C13H24	0.87	1.18	Pentadecanediol	0.00	1.18
C14H28	2.24	2.57	Pentadecanol	0.02	2.56
C15H30	1.87	2.13	Hexadecanol	0.00	2.13
C16H32	1.62	1.83	Heptadecanol	0.00	1.83
C17H34	1.24	1.41	Octadecanol	0.00	1.41
C18H36	1.24	1.41	Nonadecanol	0.00	1.41
C19H38	1.00	1.11	Eicosanol	0.00	1.11
BENZENE	2.11	2.11	Benzene	2.11	0.00
TOLUENE	3.11	3.11	Toluene	3.11	0.00
XYLENE	3.11	3.11	Xylene	3.11	0.00
C9ARO	0.75	0.75	C9Mono-Aromatic	0.75	0.00
C10ARO	0.87	0.87	C10Mono-Aromatic	0.87	0.00
C11ARO	1.37	1.37	C11Mono-Aromatic	1.37	0.00
C12ARO	0.50	0.50	C12Mono-Aromatic	0.50	0.00
C13DIARO	0.75	0.75	C13Dicyclo-Aromatic	0.75	0.00
C15DIARO	0.75	0.75	C15Dicyclo-Aromatic	0.75	0.00

Table S9. Summary of materials balance of pyrolysis oil separation (**Fig. S17B, Pyrolysis Oil Pretreatment**) simulation using Aspen Plus V12.1. Precursors for di-alcohol are highlighted in blue. ARO represents monoaromatic species and DIARO represents diaromatic species.

Species	Mass Flow (kg/h)	Stream								Corresponding Final Product	Boiling Point Corresponding Final Product (°C)
		S1 (kg/h)	S1 Split Fraction	S2 (kg/h)	S2 Split Fraction	S3 (kg/h)	S3 Split Fraction	S4 (kg/h)	S4 Split Fraction		
Mass Flows	100.00	55.25		13.00		15.00		17.00			
C5H12	5.97	5.90	98.80%	0.07	1.10%	0.01	0.10%	0.00	0.00%	Pentane	36
C5H10	6.09	6.05	99.21%	0.04	0.74%	0.00	0.05%	0.00	0.00%	Hexanol	156
C5H8	3.11	3.09	99.32%	0.02	0.64%	0.00	0.04%	0.00	0.00%	Heptanediol	250
C6H14	6.59	6.24	94.64%	0.29	4.35%	0.07	1.01%	0.00	0.00%	Hexane	69
C6H12	6.09	5.83	95.66%	0.22	3.62%	0.04	0.73%	0.00	0.00%	Heptanol	176
C6H10	3.86	3.71	96.29%	0.12	3.12%	0.02	0.59%	0.00	0.00%	Octanediol	279
C7H16	0.87	0.73	84.24%	0.10	11.72%	0.04	4.04%	0.00	0.00%	Heptane	98
C7H14	7.34	6.33	86.32%	0.75	10.25%	0.25	3.43%	0.00	0.00%	Octanol	195
C7H12	3.86	3.40	88.28%	0.34	8.89%	0.11	2.83%	0.00	0.00%	Nonanediol	288
C8H18	1.00	0.62	62.12%	0.27	27.21%	0.11	10.61%	0.00	0.06%	Octane	126
C8H16	4.85	3.25	67.05%	1.15	23.74%	0.45	9.17%	0.00	0.04%	Nonanol	214
C8H14	3.11	2.21	71.16%	0.65	20.85%	0.25	7.97%	0.00	0.03%	Decanediol	297
C9H20	0.37	0.07	18.32%	0.21	56.87%	0.09	24.01%	0.00	0.80%	Nonane	151
C9H18	4.10	1.06	25.72%	2.16	52.56%	0.87	21.18%	0.02	0.54%	Decanol	231
C9H16	2.24	0.85	37.82%	1.00	44.47%	0.39	17.42%	0.01	0.29%	Undecanediol	321
C10H22	0.50	0.00	0.19%	0.17	34.86%	0.28	56.13%	0.04	8.81%	Decane	174
C10H20	4.35	0.02	0.48%	1.88	43.17%	2.17	49.88%	0.28	6.47%	Undecanol	247
C10H18	1.74	0.03	1.74%	0.93	53.68%	0.71	40.80%	0.07	3.78%	Dodecanediol	340
C11H22	3.48	0.00	0.00%	0.09	2.61%	2.41	69.28%	0.98	28.11%	Tridecanol	264
C11H20	1.12	0.00	0.00%	0.06	5.71%	0.79	71.00%	0.26	23.29%	Tridecanediol	347
C12H24	2.74	0.00	0.00%	0.00	0.02%	1.27	46.57%	1.46	53.42%	Dodecanol	280
C12H22	0.87	0.00	0.00%	0.00	0.17%	0.50	57.26%	0.37	42.57%	Tetradecanediol	356
C13H26	2.36	0.00	0.00%	0.00	0.00%	0.60	25.46%	1.76	74.54%	Tetradecanol	294
C13H24	0.87	0.00	0.00%	0.00	0.00%	0.25	28.67%	0.62	71.33%	Pentadecanediol	366
C14H28	2.24	0.00	0.00%	0.00	0.00%	0.26	11.61%	1.98	88.39%	Pentadecanol	308
C15H30	1.87	0.00	0.00%	0.00	0.00%	0.08	4.53%	1.78	95.47%	Hexadecanol	325
C16H32	1.62	0.00	0.00%	0.00	0.00%	0.03	1.59%	1.59	98.41%	Heptadecanol	338
C17H34	1.24	0.00	0.00%	0.00	0.00%	0.01	0.57%	1.24	99.43%	Octadecanol	350
C18H36	1.24	0.00	0.00%	0.00	0.00%	0.00	0.21%	1.24	99.79%	Nonadecanol	362
C19H38	1.00	0.00	0.00%	0.00	0.00%	0.00	0.06%	0.99	99.94%	Eicosanol	372
BENZENE	2.11	1.95	92.11%	0.13	6.22%	0.04	1.67%	0.00	0.00%	Benzene	80
TOLUENE	3.11	2.40	77.28%	0.52	16.68%	0.19	6.03%	0.00	0.01%	Toluene	110
XYLENE	3.11	1.30	41.90%	1.30	41.75%	0.50	16.11%	0.01	0.23%	Xylene	138
C9ARO	0.75	0.01	1.84%	0.40	54.07%	0.30	40.41%	0.03	3.68%	C9Mono-Aromatic	169

C10ARO	0.87	0.00	0.00%	0.07	7.87%	0.62	70.96%	0.18	21.16%	C10Mono-Aromatic	187
C11ARO	1.37	0.00	0.00%	0.00	0.36%	0.83	60.86%	0.53	38.78%	C11Mono-Aromatic	201
C12ARO	0.50	0.00	0.00%	0.00	0.00%	0.20	40.65%	0.30	59.34%	C12Mono-Aromatic	237
C13DIARO	0.75	0.00	0.00%	0.00	0.00%	0.16	22.00%	0.58	78.00%	C13Dicyclo-Aromatic	218
C15DIARO	0.75	0.00	0.00%	0.00	0.00%	0.08	10.44%	0.67	89.56%	C15Dicyclo-Aromatic	258

Table S10. Stream data for the separation of final products (**Fig. S17B, Separation**) simulation using Aspen Plus V12.1. Di-alcohols are highlighted in blue.

Pyrolysis Oil	S1 (kg/h)	Corresponding Final Products	Corresponding Final Products (kg/h)	S5 (kg/h)	S6 (kg/h)	S7 (kg/h)	S8 (kg/h)	S9 (kg/h)	S10 (kg/h)	S11 (kg/h)	S12 (kg/h)
Sum	55.25		73.09	53.63	19.46	22.08	31.56	6.18	25.38	3.93	15.53
C5H12	5.90	Pentane	5.90	5.898	0.000	5.898	0.000	0.00	0.00	0.00	0.00
C5H10	6.05	Hexanol	8.83	8.828	0.000	2.343	6.485	5.63	0.85	0.00	0.00
C5H8	3.09	Heptanediol	5.99	2.002	3.986	0.000	2.002	0.00	2.00	3.91	0.08
C6H14	6.24	Hexane	6.24	6.238	0.000	6.238	0.000	0.00	0.00	0.00	0.00
C6H12	5.83	Heptanol	8.05	8.048	0.000	0.533	7.515	0.44	7.08	0.00	0.00
C6H10	3.71	Octanediol	6.61	1.609	5.000	0.000	1.609	0.00	1.61	0.01	4.99
C7H16	0.73	Heptane	0.73	0.730	0.000	0.730	0.000	0.00	0.00	0.00	0.00
C7H14	6.33	Octanol	8.43	8.428	0.000	0.112	8.316	0.00	8.32	0.00	0.00
C7H12	3.40	Nonanediol	5.68	0.005	5.674	0.000	0.005	0.00	0.00	0.00	5.67
C8H18	0.62	Octane	0.62	0.620	0.000	0.616	0.004	0.00	0.00	0.00	0.00
C8H16	3.25	Nonanol	4.20	4.199	0.000	0.000	4.199	0.00	4.20	0.00	0.00
C8H14	2.21	Decanediol	3.50	0.000	3.499	0.000	0.000	0.00	0.00	0.00	3.50
C9H20	0.07	Nonane	0.07	0.070	0.000	0.023	0.047	0.04	0.01	0.00	0.00
C9H18	1.06	Decanol	1.32	1.309	0.010	0.000	1.309	0.00	1.31	0.01	0.00
C9H16	0.85	Undecanediol	1.29	0.000	1.290	0.000	0.000	0.00	0.00	0.00	1.29
BENZENE	1.95	Benzene	1.95	1.949	0.000	1.949	0.000	0.00	0.00	0.00	0.00
TOLUENE	2.40	Toluene	2.40	2.399	0.000	2.399	0.000	0.00	0.00	0.00	0.00
XYLENE	1.30	Xylene	1.30	1.300	0.000	1.235	0.065	0.07	0.00	0.00	0.00

Table S11. Carbon emissions from the production of chemicals using traditional methods

Chemicals	Carbon emission (kg eCO ₂ / kg of chemical)
Mono-Aromatics	1.932
Dicyclo-Aromatics	0.524
C5-C8 Paraffins	0.965
>C8 Paraffins	0.524
Alcohols	2.724
Dialcohols	15.150
Hemiacetal	1.655

We used the openLCA v1.10.335 (32) software to perform Life Cycle Assessment (LCA) calculations with the Environmental Footprint (EF 2.0) as the impact assessment method, evaluating all products within the system. We defined the functional unit as processing one ton of plastic waste when assessing the carbon emissions of the proposed technology route. The mass allocation method was used to distribute the carbon emissions among the chemicals produced within the system.

Table S12. Assumptions and parameters for the economic evaluation.(66-69)

Income tax [%]	39%
Interest rate [%]	10%
Lifetime [years]	20
Operating time [hours/year]	350
Plant depreciation [years]	10
Labor cost [\$ /year]	400,000
Maintenances [\$ /year]	3% of CAPEX
Insurance [\$ /year]	0.7% of CAPEX
Hydrogen [\$ /ton]	1507
Syngas [\$ /ton]	536

Table S13. Cost and carbon emissions of technologies. (13, 70, 71)

Technology	Input Material	OPEX [\$ /ton]	Scale [kton/year]	CAPEX [\$MM]	Emission [kg eCO ₂ / kg feed]
Pyrolysis	Plastic Waste	30.0	36.5	38.1	0.436
Hydroformylation	Pyrolysis Oil	588.0	214.1	32.0	0.467
Hydrogenation	Hydroformylated Oil	34.0	21.3	22.5	0.520
Pyrolysis-Oil Separation	Pyrolysis Oil	3.8	500.0	6.6	0.012
Products Separation	Hydrotreated Oil	11.0	120.0	3.9	0.045

Table S13 presents the capital expenditures (CAPEX), operating expenditures (OPEX) and the carbon emissions of technologies within the system. The so called sixth scaling rule was used to account for economies of scale when evaluating CAPEX of facilities:

$$\alpha^t = \bar{\alpha}^t \cdot \left(\frac{\xi^t}{\bar{\xi}^t} \right)^\beta$$

Here, α represents CAPEX of technology t at a scale ξ . $\bar{\xi}$ is the CAPEX of such technology at base case scale $\bar{\alpha}$, and β is the scaling factor, which assumed to be 0.6 in this work.

Table S14. Ratio of the 1-octanol blobs volume to branched C8 alcohols volume in GC×GC of different upgraded plastic pyrolysis oil.

Feedstock plastic	1-octanol blob volume/ branched C8 alcohols blobs volume
virgin HDPE	38%
virgin LDPE	45%
virgin PP	2%

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